

Unit 4 — How bad would 2026-XJ actually be?

Work · kinetic and potential energy · conservation · springs · power · TNT yield · escape velocity

◆ NASA / PLANETARY DEFENSE BRIEFING — UPDATE 4

The Unit 3 DART feasibility analysis came back INCONCLUSIVE. The required Δv depends sensitively on lead time — and we may not have lead time. So now Unit 4 has to answer the OTHER branch: if 2026-XJ does impact, HOW BAD IS IT ACTUALLY?

Reports talk about "Tunguska-scale" or "Chicxulub-scale" events — but those are TNT-equivalent yields. To compute a yield, we need a framework we don't have yet: ENERGY. Specifically, the asteroid's KINETIC ENERGY at impact, converted into the same units the public uses for nuclear weapons and historical impacts.

Unit 4 builds that framework. Day 19 is the payoff — we put a number on 2026-XJ and place it on the same scale as Hiroshima, Tunguska, Chelyabinsk, and Chicxulub.

And there's a parallel question we owe ourselves: could 2026-XJ have ESCAPED — slipped past Earth and out of the system — if it had been moving faster, or if we'd pushed it sooner? That's escape velocity. The same energy framework answers both questions.

This is fiction. The physics — and the equations — are real.

MA Standards: HS-PS3-1, HS-PS3-2, HS-PS3-3, HS-PS3-4, HS-ETS1-2

22 meetings · Fall 2026 (Dec 2 – Jan 16)

Name: _____ Class period: _____ Date
received: _____

"Unit 3 gave us force. Unit 4 gives us damage — in the units the world uses."

Table of Contents

Day-type key: A = Anchor · S = Standard · L = Lab · T = Transfer

Section	Type	Page
Cover	—	1
Table of Contents	—	2
Phase 1 — Re-anchor: the impact answer needs energy	—	
Day 1 — Re-anchoring: damage isn't just force	A	3
Day 2 — Work: $W = F \cdot d$	S	6
Day 3 — Work at an angle + negative work	S	9
Phase 2 — Kinetic energy	—	
Day 4 — Kinetic energy: $KE = \frac{1}{2}mv^2$	S	12
Day 5 — Work-energy theorem: worked examples	S	15
Day 6 — Vernier KE lab	L	18
Phase 3 — Potential energy	—	
Day 7 — Gravitational PE near Earth: $U = mgh$	S	22
Day 8 — When mgh fails: g changes with altitude	S	25
Day 9 — Universal PE: $U = -GMm/r$	S	28
Phase 4 — Conservation of mechanical energy	—	
Day 10 — Conservation: $KE + PE = \text{const}$	S	31
Day 11 — Rolling, sliding, swinging	S	34
Day 12 — Roller coaster + pendulum	S	37
Day 13 — Vernier ramp lab	L	40
Phase 5 — Springs and power	—	
Day 14 — Hooke's law: $F = -kx$	S	44
Day 15 — Spring PE: $\frac{1}{2}kx^2$	S	47
Day 16 — Power: $P = W/t = F \cdot v$	S	50
Phase 6 — Energy in collisions revisited	—	
Day 17 — KE in elastic vs. inelastic collisions	S	53
Day 18 — Heat as the missing energy (Unit 5 bridge)	S	56
Phase 7 — Asteroid payoff	—	

Day 19 — TNT yield: 2026-XJ on the impact scale	S★	59
Day 20 — Escape velocity	S	62
<i>Phase 8 — Composition & assessment</i>	—	
Day 21 — Three-branch energy budget	S	65
Day 22 — Unit 4 transfer task	T	69
Equation Reference Card	—	72
Mastery Growth Chart	—	74

Day 1 — Re-anchoring — Damage Isn't Just Force

Today's Target

I can articulate why MOMENTUM gave us force at impact, but didn't give us a DAMAGE answer — and I can predict that something involving SPEED-SQUARED is doing the work.

ASTEROID THREAD

What we know about 2026-XJ today: From Unit 3: 2026-XJ's momentum at impact is enormous, and the AVERAGE force of impact is $\sim 10^{14}$ N. But "force" doesn't tell us crater size, fire radius, or tsunami height.

How today connects: What people quote for asteroid impacts is the asteroid's KINETIC ENERGY in TNT-equivalents. Unit 4 builds that framework. Day 19 cashes it in.

NOTICE AND WONDER

Watch the drop demo: tennis ball from waist, tennis ball from head, baseball from head. Rank "how hard did the floor get hit" 1-2-3. What did you NOTICE? What do you WONDER?

I notice _____

I wonder _____

DOODLE ZONE

Rank the three drops 1-2-3. What stays the same A vs. B? What changes? Same question for B vs. C. What TWO things matter for damage?

Drop test: rank how hard the floor was hit.

A — TENNIS BALL · waist height **B — TENNIS BALL · head height** **C — BASEBALL · head height**



A vs B (same mass, different SPEED):
which matters more?

B vs C (same SPEED, different mass):
which matters more?

Guess: damage $\sim m \cdot v$?
or $m \cdot v^2$?

MY GUESS — DOES DAMAGE GROW LIKE $m \cdot v$, OR FASTER?

Spoiler: the answer is KINETIC ENERGY = $\frac{1}{2} \cdot m \cdot v^2$ — and that SQUARED v is the load-bearing feature of the entire unit. We don't formalize it today; we feel it.

EXIT TICKET — DAY 1

If you DOUBLE a moving car's SPEED, does the damage in a crash double, or more than double? Why might that be? (One sentence, your best guess.)

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can explain why momentum isn't a damage answer.	☐	☐	☐	☐
I can predict that doubling speed gives MORE than double the damage.	☐	☐	☐	☐
I can name the two questions Unit 4 must answer (impact yield, escape).	☐	☐	☐	☐

Day 2 — Work: $W = F \cdot d$

Today's Target

I can write $W = F \cdot d$, give units (Joules = $\text{N} \cdot \text{m}$), compute work in 1D, and identify cases where work is ZERO even though a force is applied.

ASTEROID THREAD

What we know about 2026-XJ today: Earth's gravity does WORK on 2026-XJ as it falls. That work is what increases the asteroid's energy.

How today connects: Work is the bridge between FORCE (Unit 1, Unit 3) and ENERGY (today on). It's how energy MOVES from one object to another.

VOCAB · WORK (W)

Definition: Energy transferred to (or from) an object via a force applied along its motion. $W = F \cdot d$.

Example: Pushing a 50-N box 4 m $\rightarrow W = 200 \text{ J}$. Lifting a 20-N book 1.5 m $\rightarrow W = 30 \text{ J}$.

Cognates: *Sp. trabajo · Pt. trabalho · HC travay*

Watch out: Work is NOT the same as effort. If you push a wall and it doesn't move, you've done ZERO work — even though you got tired. (Your body did work, but not on the wall.)

VOCAB · JOULE (J)

Definition: The unit of work and energy. 1 Joule = 1 Newton · 1 meter.

Example: Lifting an apple ($\sim 1 \text{ N}$) up 1 m takes about 1 J of work.

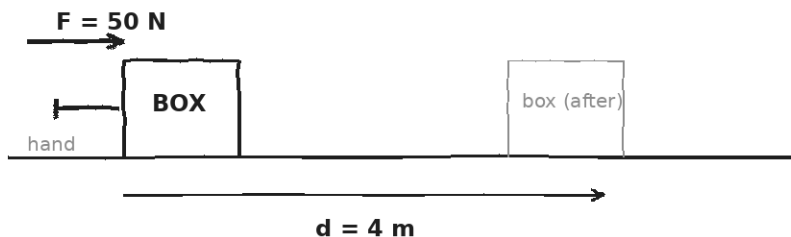
Cognates: *Sp. julio · Pt. joule · HC joul*

Watch out: A Joule is small. A snack bar has $\sim 10^6 \text{ J}$. The asteroid's KE is $\sim 10^{17} \text{ J}$. You'll see scientific notation a lot this unit.

DOODLE ZONE

Compute W for the pushed box. Then write zero for each zero-work case and **EXPLAIN** why (one phrase each).

Push a box across the floor. $W = F \cdot d$



Compute the work:

$$W = F \cdot d$$

$$W = 50\text{ N} \cdot 4\text{ m}$$

$$W = \underline{\hspace{2cm}}\text{ J}$$

Zero-work cases:

- push wall, $d = 0$

$$W = \underline{\hspace{2cm}}$$

- carry book level

$$W = \underline{\hspace{2cm}}$$

WORKED EXAMPLE — PUSHING A BOX

You push a 50-N box 4 m across the floor with a constant horizontal force. How much work do you do?

GIVEN	$F = 50\text{ N (along motion)} \cdot d = 4\text{ m}$
EQUATION	$W = F \cdot d$
WORK	$W = 50 \cdot 4$ $W = 200\text{ J}$
ANSWER	200 J of work done on the box.

YOUR TURN — LIFTING A BOOK

You lift a 20-N book straight up 1.5 m at constant speed. How much work do you do AGAINST gravity?

GIVEN	
EQUATION	
WORK	
ANSWER	

YOUR TURN — THREE ZERO-WORK CASES

For each case, write W and explain in ONE phrase why it's zero. (a) You push hard on a wall; the wall doesn't move. (b) You carry a 20-N book level across a 5-m room. (c) A satellite in CIRCULAR orbit moves around Earth.

GIVEN	
EQUATION	
WORK	
ANSWER	

	Work needs BOTH a force AND motion in the direction of that force. Lose either one $\rightarrow W = 0$.
---	--

Fill in the frame:

Carrying a book level across the room does NO WORK on the book because the force is _____ and the motion is _____ — and the angle between them is _____°.

EXIT TICKET — DAY 2

You lift a 20-N book 1.5 m, then walk 3 m holding the book at that height. How much work did you do TOTAL on the book?

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Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute $W = F \cdot d$ with correct units.	☐	☐	☐	☐
I can identify zero-work cases and explain why.	☐	☐	☐	☐
I can compute work done against gravity when lifting.	☐	☐	☐	☐

Day 3 — Work at an Angle + Negative Work

Today's Target

I can compute $W = F \cdot d \cdot \cos \theta$ for an angled force, and I can assign the correct SIGN of work for forces that aid or oppose motion.

ASTEROID THREAD

What we know about 2026-XJ today: If 2026-XJ enters our atmosphere, air drag does NEGATIVE work on it — taking energy AWAY.

How today connects: That's why meteors GLOW. Their KE is converting to heat and light because atmospheric friction did massive negative work.

VOCAB · WORK AT AN ANGLE

Definition: When the force F makes an angle θ with the direction of motion d , only the COMPONENT of F along d does work: $W = F \cdot d \cdot \cos \theta$.

Example: Pulling a suitcase with a 100-N handle force at 30° above horizontal over 10 m: $W = 100 \cdot 10 \cdot \cos 30^\circ \approx 866 \text{ J}$.

Cognates: *Sp. trabajo con ángulo · Pt. trabalho com ângulo · HC travay ak ang*

Watch out: At $\theta = 90^\circ$, $\cos \theta = 0$ and $W = 0$ — even though there's a big force.

VOCAB · NEGATIVE WORK

Definition: Work done by a force that opposes the motion (θ between 90° and 180°). $\cos \theta$ is negative, so W is negative — energy is taken AWAY from the object.

Example: Friction always does negative work on a sliding object. $\cos 180^\circ = -1$.

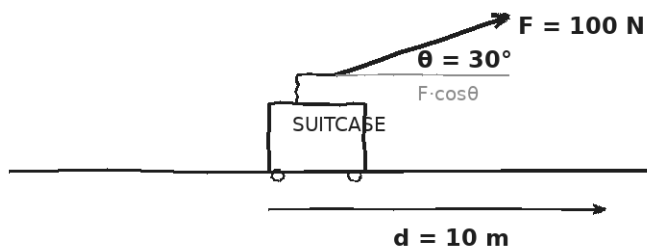
Cognates: *Sp. trabajo negativo · Pt. trabalho negativo · HC travay negatif*

Watch out: "Negative work" doesn't mean "no work." It means energy is being REMOVED, not added.

DOODLE ZONE

Compute the work done by the 100-N pull. Then compute the friction's work. Net?

Pull a suitcase at an angle. $W = F \cdot d \cdot \cos \theta$



Compute W:

$$W = F \cdot d \cdot \cos \theta$$

$$W = 100 \cdot 10 \cdot \cos 30^\circ$$

$$= 1000 \cdot 0.866$$

$$= \underline{\hspace{1cm}} \text{ J}$$

If friction = 20 N

opposes motion:

$$W_{\text{fric}} = 20 \cdot 10 \cdot \cos 180^\circ$$

$$= \underline{\hspace{1cm}} \text{ J}$$

(negative work!)

WORKED EXAMPLE — PULLING A SUITCASE AT ANGLE

You pull a 100-N handle force on a suitcase 10 m at 30° above the horizontal. Friction opposes the motion with 20 N. Find: (a) work done by you, (b) work done by friction, (c) NET work.

GIVEN	$F_{\text{you}} = 100 \text{ N}$, $\theta_{\text{you}} = 30^\circ$ · $F_{\text{fric}} = 20 \text{ N}$, $\theta_{\text{fric}} = 180^\circ$ $d = 10 \text{ m}$
EQUATION	$W = F \cdot d \cdot \cos \theta$ (for each force)
WORK	$W_{\text{you}} = 100 \cdot 10 \cdot \cos 30^\circ = 1000 \cdot 0.866 = 866 \text{ J}$ $W_{\text{fric}} = 20 \cdot 10 \cdot \cos 180^\circ = 200 \cdot (-1) = -200 \text{ J}$ $W_{\text{net}} = 866 + (-200) = 666 \text{ J}$
ANSWER	You did 866 J of POSITIVE work. Friction did 200 J of NEGATIVE work. Net: 666 J.

YOUR TURN — PUSH ACROSS THE FLOOR

You push a 30-N box 4 m across the floor at constant speed against 12 N of friction. (a) What work did YOU do? (b) What work did friction do? (c) What's the NET work?

GIVEN	
EQUATION	
WORK	
ANSWER	

	ALWAYS write the angle BEFORE you compute. θ between F and d, NOT between F and the floor. $\cos 180^\circ = -1$, $\cos 90^\circ = 0$, $\cos 0^\circ = 1$.
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Fill in the frame:

Friction does NEGATIVE work because the friction force points

_____ to the motion, so $\cos \theta =$

_____, so W comes out

_____.

EXIT TICKET — DAY 3

You push a 30-N box 4 m at constant speed against 12 N of friction. What's the NET work?

TODAY'S TARGETS — RATE YOURSELF

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Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply $W = F \cdot d \cdot \cos \theta$ for an angled force.	☐	☐	☐	☐
I can assign the correct sign to work done by aiding vs. opposing forces.	☐	☐	☐	☐
I can sum positive and negative works to find NET work.	☐	☐	☐	☐

Day 4 — Kinetic Energy + the Squared-v Reveal

Today's Target

I can write $KE = \frac{1}{2}mv^2$, compute KE for moving objects across mass and speed scales, and state the work-energy theorem $W_{\text{net}} = \Delta KE$.

ASTEROID THREAD

What we know about 2026-XJ today: The DAMAGE number for asteroids is their KINETIC ENERGY at impact, not their momentum.

How today connects: Today's equation is the heart of Unit 4. By Day 19, we plug 2026-XJ into it and get its yield in tons of TNT.

VOCAB · KINETIC ENERGY (KE)

Definition: The energy of motion. $KE = \frac{1}{2} \cdot m \cdot v^2$. Units: Joules ($\text{kg} \cdot \text{m}^2/\text{s}^2$). Always positive, regardless of direction.

Example: A 1500-kg car at 30 m/s has $KE = \frac{1}{2} \cdot 1500 \cdot 900 = 675,000 \text{ J}$. At 60 m/s: $KE = 2,700,000 \text{ J}$ — FOUR times as much.

Cognates: *Sp. energía cinética · Pt. energia cinética · HC enèji sinetik*

Watch out: The squared v matters. KE is NOT proportional to v — it's proportional to v^2 . Doubling speed quadruples KE.

VOCAB · WORK-ENERGY THEOREM

Definition: The NET work done on an object equals the change in its KE: $W_{\text{net}} = \Delta KE = KE_f - KE_i$.

Example: A 1-kg cart at rest gets 50 J of net work $\rightarrow KE_f = 50 \text{ J} \rightarrow v_f = \sqrt{(2 \cdot 50/1)} = 10 \text{ m/s}$.

Cognates: *Sp. teorema del trabajo-energía · Pt. teorema do trabalho-energia*

Watch out: It's the NET work that matters — sum of ALL forces' contributions, not just one.

DOODLE ZONE

Compute KE at each speed using $\frac{1}{2}mv^2$. Confirm: doubling v gives $4 \times KE$; tripling v gives $9 \times$.

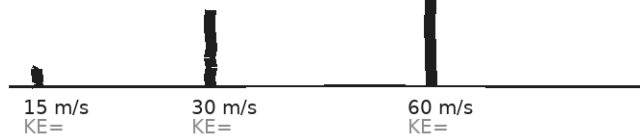
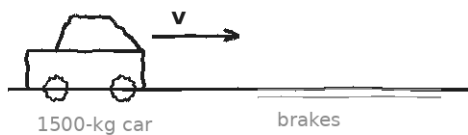
$KE = \frac{1}{2} m v^2$. Double the speed $\rightarrow 4 \times$ the KE .

KE at three speeds (1500 kg)

v doubles $\rightarrow KE \times 4$

v quadruples $\rightarrow KE \times 16$

fill in the three KE values using $\frac{1}{2}mv^2$.



WORKED EXAMPLE — KE OF A CAR AT TWO SPEEDS

A 1500-kg car moves at (a) 30 m/s, (b) 60 m/s. Find the KE in each case.

GIVEN	$m = 1500 \text{ kg} \cdot v_a = 30 \text{ m/s} \cdot v_b = 60 \text{ m/s}$
EQUATION	$KE = \frac{1}{2} \cdot m \cdot v^2$
WORK	$KE_a = \frac{1}{2} \cdot 1500 \cdot 30^2 = \frac{1}{2} \cdot 1500 \cdot 900 = 675,000 \text{ J}$ $KE_b = \frac{1}{2} \cdot 1500 \cdot 60^2 = \frac{1}{2} \cdot 1500 \cdot 3600 = 2,700,000 \text{ J}$ $KE_b / KE_a = 4$ (matches: v doubled $\rightarrow KE \times 4$)
ANSWER	At 30 m/s: $6.75 \times 10^5 \text{ J}$. At 60 m/s: $2.7 \times 10^6 \text{ J}$ — FOUR times the energy.

YOUR TURN — KE of 2026-XJ

2026-XJ has $m = 1 \times 10^9$ kg and impact speed $v = 3 \times 10^4$ m/s. Find its KE at impact.

GIVEN	
EQUATION	
WORK	
ANSWER	

YOUR TURN — WORK-ENERGY THEOREM

A 2-kg ball moves at 3 m/s. A net 27 J of work is done on it. What's its final speed? (Use $W_{\text{net}} = \Delta KE$.)

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Highway speeds aren't twice as dangerous as 30 mph. They're FOUR TIMES. This is the squared v at work — and it's why 2026-XJ matters.
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Fill in the frame:

Doubling a car's speed _____ the kinetic energy, because KE depends on v _____ (not just v).

EXIT TICKET — DAY 4

A 2-kg ball at 3 m/s has 9 J of KE. If a net 27 J of work is done on it, what's its final speed?

TODAY'S TARGETS — RATE YOURSELF

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Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute $KE = \frac{1}{2}mv^2$ with correct units.	☐	☐	☐	☐
I can predict the ratio of KEs when speed changes.	☐	☐	☐	☐
I can apply $W_{\text{net}} = \Delta KE$ to find final speed.	☐	☐	☐	☐

Day 5 — Work-Energy Theorem — Worked Examples

Today's Target

I can apply $W_{\text{net}} = \Delta KE$ as a shortcut to find stopping distance, final speed, or net force without solving for time.

ASTEROID THREAD

What we know about 2026-XJ today: Gravity does work on the falling asteroid. That work IS its impact KE.

How today connects: The work-energy theorem lets us skip time-history of the fall and go straight from forces to impact speed. That's tomorrow's lab.

VOCAB · $W_{\text{NET}} = \Delta KE$ (SHORTCUT FORM)

Definition: When forces are CONSTANT, $W_{\text{net}} = F_{\text{net}} \cdot d \cdot \cos\theta = \frac{1}{2}m \cdot v_f^2 - \frac{1}{2}m \cdot v_i^2$. Three knowns → one unknown.

Example: A 1500-kg car at 30 m/s, brakes apply 9000 N. Stopping distance? $W_{\text{net}} = -F \cdot d = 0 - \frac{1}{2}mv^2 \rightarrow d = \frac{1}{2} \cdot 1500 \cdot 900 / 9000 = 75 \text{ m}$.

Cognates: *Sp. teorema trabajo-energía aplicado · Pt. teorema aplicado*

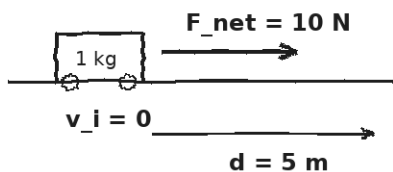
Watch out: W_{net} is the work of the NET force, not any single force. Add up ALL the works (with signs) first.

DOODLE ZONE

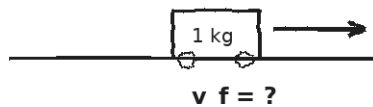
Solve for v_f using $W_{\text{net}} = \Delta KE$. Show why this is faster than kinematics.

Work-energy theorem: $W_{\text{net}} = \Delta KE$

BEFORE



AFTER



$$W_{\text{net}} = F \cdot d = (10)(5) = 50 \text{ J}$$

$$\Delta KE = \frac{1}{2}mv_f^2 - 0 = 50 \text{ J} \rightarrow v_f = \sqrt{(2 \cdot 50 / 1)} = \underline{\hspace{1cm}} \text{ m/s}$$

WORKED EXAMPLE — STOPPING DISTANCE OF A CAR

A 1500-kg car at 30 m/s brakes to rest. The brakes apply 9000 N (constant). How far does the car travel before stopping?

GIVEN	$m = 1500 \text{ kg} \cdot v_i = 30 \text{ m/s} \cdot v_f = 0 \cdot F_{\text{brake}} = 9000 \text{ N}$ (opposes motion)
EQUATION	$W_{\text{net}} = \Delta KE \rightarrow -F \cdot d = 0 - \frac{1}{2}m \cdot v_i^2$
WORK	$\begin{aligned} -9000 \cdot d &= -\frac{1}{2} \cdot 1500 \cdot 30^2 \\ -9000 \cdot d &= -675,000 \\ d &= 675,000 / 9000 = 75 \text{ m} \end{aligned}$
ANSWER	The car travels 75 m before stopping.

WORKED EXAMPLE — ROLLER-COASTER: FROM GRAVITY WORK TO BOTTOM SPEED

A 100-kg coaster cart starts at rest at the top of a 10-m incline. Friction-free. What's its speed at the bottom?

GIVEN	$m = 100 \text{ kg} \cdot v_i = 0 \cdot h = 10 \text{ m} \cdot g = 9.81 \text{ m/s}^2$
EQUATION	$W_{\text{grav}} = m \cdot g \cdot h \quad \text{AND} \quad W_{\text{net}} = \Delta KE = \frac{1}{2} m \cdot v_f^2$
WORK	$W_{\text{grav}} = 100 \cdot 9.81 \cdot 10 = 9810 \text{ J}$ $9810 = \frac{1}{2} \cdot 100 \cdot v_f^2$ $v_f^2 = 196.2$ $v_f \approx 14 \text{ m/s}$
ANSWER	$\approx 14 \text{ m/s}$ at the bottom. Faster than kinematics — no time needed.

YOUR TURN — SPRING-LAUNCHED BALL

A 0.5-kg ball is fired from a spring that does 8 J of work on it. The ball starts at rest. What's its speed leaving the spring?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	If you KNOW the forces and DISTANCES, skip kinematics. Compute the work, set it equal to ΔKE , solve.
---	---

Fill in the frame:

The work-energy theorem is faster than kinematics when I already know the

_____ and the

_____, but not the

_____.

EXIT TICKET — DAY 5

A 0.5-kg ball is fired from a spring that does 8 J of work on it. The ball starts at rest. What's its speed?

TODAY'S TARGETS — RATE YOURSELF

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Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply $W_{\text{net}} = \Delta KE$ to find stopping distance.	☐	☐	☐	☐
I can apply $W_{\text{net}} = \Delta KE$ to find final speed from gravity work.	☐	☐	☐	☐
I can choose between kinematics and the work-energy theorem.	☐	☐	☐	☐

Day 6 — Vernier KE Lab — Confirm $W_{\text{net}} = \Delta KE$ ·

Investigation 4.1

Today's Target

I can run a Vernier dynamics-cart trial with a hanging-mass pull, compute predicted v from $W_{\text{net}} = \Delta KE$, measure v with a photogate, and report the % difference with one credible source of error.

ASTEROID THREAD

What we know about 2026-XJ today: We've stated $W_{\text{net}} = \Delta KE$. Today we measure it.

How today connects: If our cart confirms the theorem, we can trust it for asteroid energy on Day 19.

LAB SETUP — INVESTIGATION 4.1

Driving question: Does the predicted final speed of a cart (from $W_{\text{net}} = \Delta KE$) match the measured speed from the photogate? Across 3 trials with different hanging masses?

Equipment:

- Vernier dynamics track (1 per pair) + 1 cart
- Pulley clamped at end of track + string
- Hanging masses (50 g, 100 g, 150 g)
- 1 photogate + LabQuest + Logger Pro or Graphical Analysis
- Balance to weigh cart (record m_{cart}) and meter stick for d

What you do:

1. Level the track. Weigh the cart and record m_{cart} .
2. Set up pulley + string. Hang first mass m_h . Compute $F_{\text{pull}} = m_h \cdot g$.
3. Mark the cart's starting position. Measure d to the photogate.
4. Release. Photogate reads v_{measured} .
5. Compute $W_{\text{net}} = F_{\text{pull}} \cdot d$, predict v from $W_{\text{net}} = \frac{1}{2}m_{\text{cart}} \cdot v^2$. Compare to measured.
6. Repeat with the other two hanging masses. Three trials total. Compute % difference.

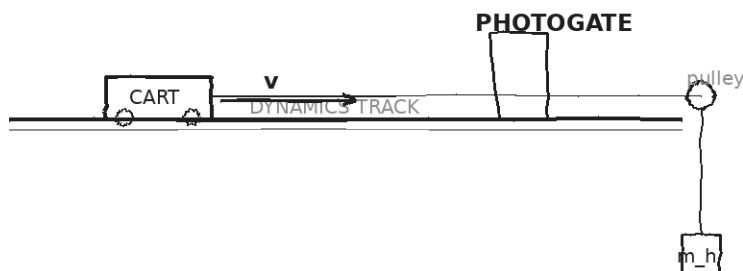
Safety: Hanging masses below knee level. Catch them at end of run. Photogates clamped firmly.

How this lab serves the year's question: If the theorem holds in our carts, the Day 19 yield calculation is on solid ground.

DOODLE ZONE

Sketch your setup. Record m_{cart} and m_h on the diagram. Fill $v_{\text{predicted}}$ and v_{measured} as you go.

Vernier KE lab: cart + pulley + hanging mass + photogate



For each trial:

$$F_{\text{pull}} = m_h \cdot g$$

$$W_{\text{net}} = F \cdot d$$

predict v from

$$W = \frac{1}{2} m_{\text{cart}} \cdot v^2$$

measure v from
photogate.

% diff = ?
(record below)

DATA — THREE TRIALS

Trial	m_h (kg)	F_{pull} (N)	d (m)	W_{net} (J)	v_{pred} (m/s)	v_{meas} (m/s)	% diff
1							
2							
3							

OBSERVATION & ANALYSIS

Did $v_{\text{predicted}}$ match v_{measured} ? Across all 3 trials, within what %?

**If there's a gap, what physical thing did NEGATIVE work that we didn't count?
Is the gap bigger for smaller or larger hanging masses?**

My v values matched within _____ %. The most likely source of error was _____, because it would _____ the cart's KE.

Asteroid connection: how does this serve the year's question?

Fill in the frame:

$W_{\text{net}} = \Delta KE$ predicts v from $W = \frac{1}{2}m \cdot v^2$. My trials confirmed this within ____ %.

The biggest source of error was

_____.

EXIT TICKET — DAY 6

Write a one-sentence CLAIM about what the lab showed, then one sentence of EVIDENCE (your numbers), then one sentence of REASONING (why the % difference is small).

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can set up a Vernier dynamics-cart + pulley + photogate.	☐	☐	☐	☐
I can predict v from $W_{\text{net}} = \Delta KE$.	☐	☐	☐	☐
I can compute % difference and name a credible source of error.	☐	☐	☐	☐

Day 7 — Gravitational PE near Earth: $U = mgh$

Today's Target

I can write $U = mgh$, compute PE relative to a chosen reference, and explain that only DIFFERENCES in PE are physically meaningful.

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ has gravitational PE relative to Earth. As it falls, that PE converts to KE.

How today connects: Near the surface, $U = mgh$ is the cleanest formula. By Day 9, we'll meet its more general cousin for asteroid distances.

VOCAB · GRAVITATIONAL PE (NEAR SURFACE)

Definition: The energy stored by lifting a mass m against gravity to height h above a reference level. $U = m \cdot g \cdot h$.

Example: A 2-kg book on a 1-m desk: $U = 2 \cdot 9.81 \cdot 1 \approx 20 \text{ J}$ (above floor). Same book on the floor: $U = 0$ (relative to floor).

Cognates: *Sp. energía potencial gravitatoria · Pt. energia potencial gravitacional · HC enèji potansyèl gravite*

Watch out: PE depends on the CHOICE of zero height. Two answers can both be "right" if they use different references.

VOCAB · REFERENCE HEIGHT

Definition: The chosen zero level for PE. Usually the floor, ground, or some convenient surface.

Example: A book on a desk has PE = 20 J relative to floor, but PE = 0 relative to the desktop.

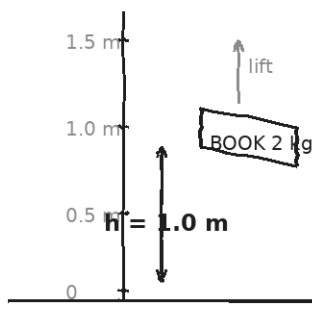
Cognates: *Sp. altura de referencia · Pt. altura de referência*

Watch out: Only DIFFERENCES in PE matter physically. The choice of zero is for our convenience.

DOODLE ZONE

Compute U at $h = 1.0$ m and $h = 1.5$ m. Then ΔU when lifting from 1.0 to 1.5.

Gravitational PE near the surface: $U = m \cdot g \cdot h$



$$U = m \cdot g \cdot h$$

($g = 9.81$ m/s², near Earth)

At $h = 1.0$ m:

$$U = 2 \cdot 9.81 \cdot 1.0 = \underline{\hspace{1cm}} \text{ J}$$

At $h = 1.5$ m:

$$U = 2 \cdot 9.81 \cdot 1.5 = \underline{\hspace{1cm}} \text{ J}$$

ΔU (lifting from

$$1.0 \text{ to } 1.5 \text{ m}) = \underline{\hspace{1cm}} \text{ J}$$

= work YOU did.

WORKED EXAMPLE — BOOK ON A DESK

A 2-kg book sits on a desk 1.0 m above the floor. Find its PE (a) relative to the floor, (b) relative to the desktop.

GIVEN	$m = 2$ kg $\cdot$ $g = 9.81$ m/s ² $\cdot$ $h_{\text{floor}} = 1.0$ m $\cdot$ $h_{\text{desk}} = 0$
EQUATION	$U = m \cdot g \cdot h$
WORK	$U_{\text{floor}} = 2 \cdot 9.81 \cdot 1.0 = 19.6$ J $U_{\text{desk}} = 2 \cdot 9.81 \cdot 0 = 0$ J
ANSWER	Both answers are correct — they just use different zero levels.

YOUR TURN — PERSON ON TOP OF A BUILDING

A 70-kg person stands at the top of a 100-m building, relative to the street. Find their PE.

GIVEN	
EQUATION	
WORK	
ANSWER	

YOUR TURN — CART ON A RAMP

A 50-kg cart sits at the top of an 8-m ramp. Find its PE relative to the bottom of the ramp.

GIVEN	
EQUATION	
WORK	
ANSWER	

	Always STATE your reference. "PE = 20 J" is meaningless without saying "above what?"
---	--

Fill in the frame:

The book on the desk has PE = 20 J above

_____, but PE = 0 above

_____. Both are correct —
 physics cares about _____.

EXIT TICKET — DAY 7
A 2-kg book on a desk 1.0 m above the floor. PE relative to floor? PE relative to desktop?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute $U = mgh$ given m , g , h .	☐	☐	☐	☐
I can choose a reference height and state it clearly.	☐	☐	☐	☐
I can explain why only ΔU is physically meaningful.	☐	☐	☐	☐

Day 8 — When mgh Fails — g changes with altitude

Today's Target

I can explain why $U = mgh$ assumes constant g and breaks at altitudes comparable to Earth's radius, and I can preview $U = -GMm/r$ as the fix.

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ is currently 40,000–100,000 km out. At that distance, g is NOT 9.81 m/s^2 .

How today connects: If we used mgh for the asteroid's PE, we'd get a hilariously wrong answer. Today we see WHY and preview the fix.

VOCAB · G VARIES WITH R

Definition: From Unit 2: $g(r) = G \cdot M / r^2$. At Earth's surface ($r \approx 6,371 \text{ km}$) $g \approx 9.81$. At 100,000 km out, $g \approx 0.04$.

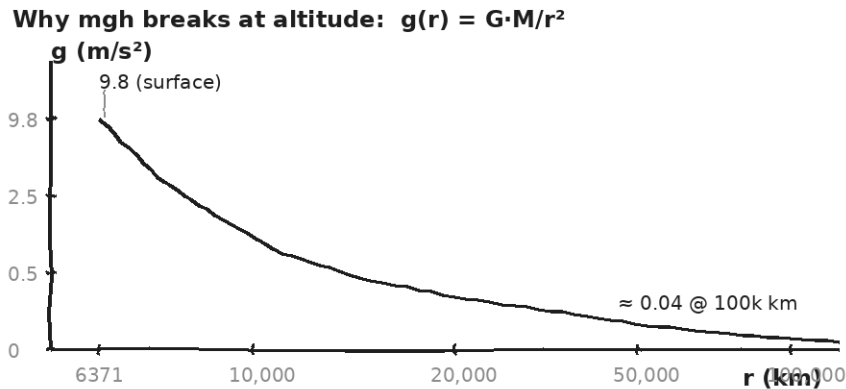
Example: The Moon ($r \approx 384,000 \text{ km}$ from Earth) feels $g_{\text{Earth}} \approx 0.003 \text{ m/s}^2$. Almost nothing.

Cognates: *Sp. g varía · Pt. g varia*

Watch out: On Earth's surface, g feels constant only because we don't move far enough vertically. Across asteroid distances, g changes by orders of magnitude.

DOODLE ZONE

On the curve, mark where $g = 9.8$ (surface). Mark where $g \approx 0.04$ (asteroid distance). What's the ratio? What does that mean for mgh ?



mgh assumes
 $g = 9.81$ always.

At asteroid
distances, g is
 $200\times$ smaller.

We need a new
formula:

$$U = -GMm/r$$

WORKED PROBE — WHAT MGH WOULD SAY AT ALTITUDE

Using $U = mgh$ with $g = 9.81 \text{ m/s}^2$ (the surface value), find the PE of a 1000-kg object at $h = 100,000 \text{ km}$ altitude. Then compute g at that altitude using $g(r) = GM/r^2$ with $r = 6.4 \times 10^6 + 1 \times 10^8 \text{ m}$. Comment on the error.

GIVEN	$m = 1000 \text{ kg} \cdot h = 1 \times 10^8 \text{ m} \cdot g_{\text{surface}} = 9.81 \text{ m/s}^2$ $G = 6.67 \times 10^{-11} \cdot M_{\text{Earth}} = 6 \times 10^{24} \text{ kg}$
EQUATION	$U_{\text{mgh}} = m \cdot g \cdot h$ AND $g(r) = G \cdot M / r^2$
WORK	$U_{\text{mgh}} = 1000 \cdot 9.81 \cdot 10^8 = 9.81 \times 10^{11} \text{ J}$ $r \approx 1.06 \times 10^8 \text{ m}$ $g(r) = (6.67 \times 10^{-11})(6 \times 10^{24}) / (1.06 \times 10^8)^2 \approx 0.036 \text{ m/s}^2$ So mgh OVERESTIMATES by a factor of $9.81 / 0.036 \approx 270\times$
ANSWER	The mgh number is 270 times too big. We need a formula that uses $g(r)$, not g_{surface} .

THE FIX — coming Day 9: $U = -G \cdot M \cdot m / r$

PE = 0 at INFINITY. PE is NEGATIVE everywhere closer (because gravity does positive work pulling things in — they LOSE potential as they fall).



The negative sign is correct and important: closer = MORE negative PE; farther = closer to zero. This lets us cleanly write $KE_{\text{at_infinity}} = 0 = KE_{\text{surface}} + U_{\text{surface}}$, which gives escape velocity.

Fill in the frame:

$U = mgh$ fails when h is comparable to

_____, because g is not

_____. We need $U =$

_____ to be accurate across

altitudes.

EXIT TICKET — DAY 8

At Earth's surface, what's the PE of a 1-kg object using $U = -GMm/r$? Compare to $U = mgh$ with $h = 0$. (Different — that's OK. Why?)

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can state the assumption behind $U = mgh$ (constant g).	☐	☐	☐	☐
I can show numerically that mgh fails at altitude.	☐	☐	☐	☐
I can preview $U = -GMm/r$ and predict its sign.	☐	☐	☐	☐

Day 9 — Universal Gravitational PE: $U = -G \cdot M \cdot m / r$

Today's Target

I can apply $U = -GMm/r$ to compute the PE of an asteroid at different distances and compute ΔU as it falls (= KE gained).

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ's PE depends on its distance from Earth. As it gets closer, U becomes MORE NEGATIVE — meaning the asteroid is GAINING kinetic energy.

How today connects: ΔU as the asteroid falls is the KE it brings to impact. Day 19 puts a number on this.

VOCAB · $U = -G \cdot M \cdot m / R$

Definition: The universal gravitational PE between two masses M and m separated by distance r . $PE = 0$ at infinity; negative everywhere closer.

Example: A 1-kg object at Earth's surface ($r = 6.37 \times 10^6$ m): $U = -(6.67 \times 10^{-11})(6 \times 10^{24})(1)/(6.37 \times 10^6) \approx -6.28 \times 10^7$ J.

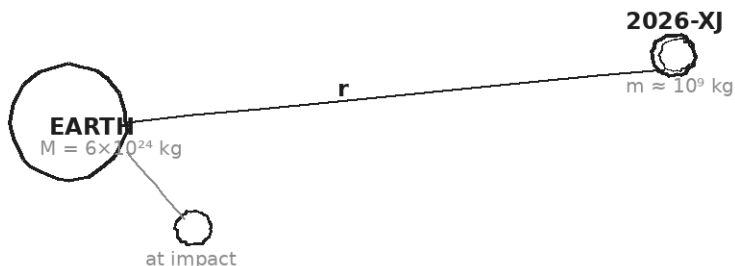
Cognates: *Sp. energía potencial gravitacional universal · Pt. energia potencial gravitacional universal*

Watch out: Negative U doesn't mean "missing energy." It means the object is in a gravitational well — you'd have to do work to lift it out to infinity.

DOODLE ZONE

Plug in numbers for U at the Moon's distance and at impact. The **DIFFERENCE** = the KE the asteroid gains as it falls.

Universal gravitational PE: $U = -G \cdot M \cdot m / r$



At $r = 384,000 \text{ km}$:

$$U_1 = -GMm/r_1 \\ = - \text{_____} \text{ J}$$

At $r = 6.4 \times 10^6 \text{ m}$:

$$U_2 = -GMm/r_2 \\ = - \text{_____} \text{ J}$$

$\Delta U = U_2 - U_1$

= (more negative)

$$= - \text{_____} \text{ J}$$

= **KE gained**
in fall.

WORKED EXAMPLE — PE OF AN ASTEROID AT THREE DISTANCES

A 1000-kg test mass at (a) 384,000 km from Earth's center, (b) Earth's surface (6,371 km), (c) at infinity. Find U at each, then ΔU from (a) to (b).

GIVEN	$m = 1000 \text{ kg} \cdot G = 6.67 \times 10^{-11} \cdot M = 5.97 \times 10^{24} \text{ kg}$ $r_a = 3.84 \times 10^8 \text{ m} \cdot r_b = 6.37 \times 10^6 \text{ m} \cdot r_c = \infty$
EQUATION	$U = -G \cdot M \cdot m / r$
WORK	$U_a = - (6.67\text{e-}11)(5.97\text{e}24)(1000)/(3.84\text{e}8) \approx -1.04 \times 10^9 \text{ J}$ $U_b = - (6.67\text{e-}11)(5.97\text{e}24)(1000)/(6.37\text{e}6) \approx -6.25 \times 10^{10} \text{ J}$ $U_c = 0 \text{ J}$ $\Delta U (a \rightarrow b) = U_b - U_a \approx -6.15 \times 10^{10} \text{ J}$
ANSWER	The 1000-kg mass GAINS $6.15 \times 10^{10} \text{ J}$ of KE falling from Moon's distance to surface.

YOUR TURN — KE GAINED FROM INFINITY

A 500-kg meteorite falls from infinity to Earth's surface (no air friction). How much KE does it gain?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	A 500-kg meteorite from infinity hits Earth at $v \approx 11.2$ km/s — Earth's escape velocity. Same number, two derivations. Day 20 confirms.
---	--

Fill in the frame:

An asteroid GAINS kinetic energy as it falls because its U becomes

_____ (more negative). The total

ME stays constant, so _____

goes up by the same amount.

EXIT TICKET — DAY 9
A 500-kg object falls from infinity to Earth's surface (no air drag). What KE does it gain?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can apply $U = -GMm/r$ at any distance.	☐	☐	☐	☐
I can compute ΔU and interpret it as KE gained.	☐	☐	☐	☐
I can predict that falling from infinity = escape-velocity KE.	☐	☐	☐	☐

Day 10 — Conservation of Mechanical Energy

Today's Target

I can state that $KE + PE = \text{constant}$ when only conservative forces act, and apply it to a simple drop.

ASTEROID THREAD

What we know about 2026-XJ today: An asteroid in free fall (no air) obeys ME conservation perfectly.

How today connects: This is the big rule of Phase 4. Days 11-13 are all special cases. Day 19 uses it to convert PE to impact KE.

VOCAB · MECHANICAL ENERGY (ME)

Definition: The sum of KE and PE: $ME = KE + PE$. When only conservative forces act (gravity, springs), ME stays constant.

Example: A ball dropped from 5 m: at top, $ME = 0 + mg \cdot 5$. At bottom, $ME = \frac{1}{2}mv^2 + 0$. Setting them equal gives v .

Cognates: *Sp. energía mecánica · Pt. energia mecânica · HC enèji mekanik*

Watch out: Friction is NOT conservative. When friction acts, ME is NOT conserved — it leaks to thermal energy.

VOCAB · CONSERVATIVE FORCE

Definition: A force whose work depends only on START and END positions, not the path. Gravity, springs. (Friction is NOT.)

Example: Walking up a ramp or a vertical lift to the same height: gravity does the SAME work.

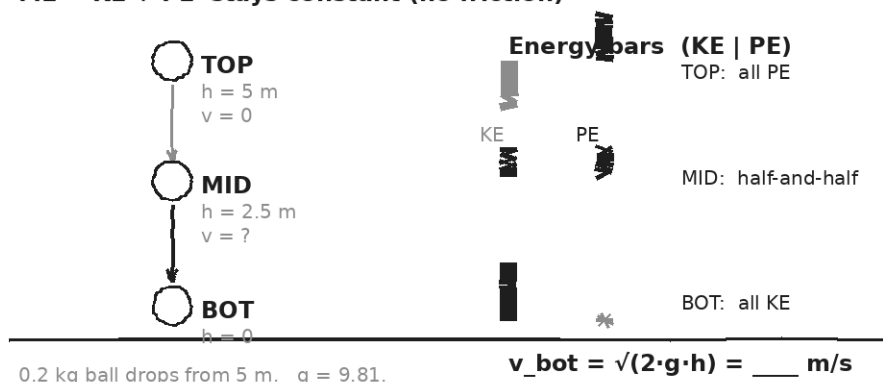
Cognates: *Sp. fuerza conservativa · Pt. força conservativa*

Watch out: "Conservative" is a math word here. It doesn't mean "saves" or "preserves." It means the work is path-independent.

DOODLE ZONE

Fill the KE / PE bars at top, middle, bottom. Their SUM should be the same at every level. Solve for v at the bottom.

ME = KE + PE stays constant (no friction)



WORKED EXAMPLE — BALL DROPPED FROM 5 M

A 0.2-kg ball is dropped (from rest) from 5 m above the floor. What's its speed at the floor?

GIVEN	$m = 0.2 \text{ kg} \cdot v_{\text{top}} = 0 \cdot h = 5 \text{ m} \cdot g = 9.81$
EQUATION	$\text{KE}_{\text{top}} + \text{PE}_{\text{top}} = \text{KE}_{\text{bot}} + \text{PE}_{\text{bot}}$ $0 + m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2 + 0$
WORK	$m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2 \quad (m \text{ cancels})$ $v = \sqrt{(2 \cdot g \cdot h)} = \sqrt{(2 \cdot 9.81 \cdot 5)}$ $v = \sqrt{98.1} \approx 9.9 \text{ m/s}$
ANSWER	$\approx 9.9 \text{ m/s}$ at impact. Mass canceled — the answer only depends on h .

YOUR TURN — DROP FROM 1.8 M

A 0.5-kg ball is dropped from 1.8 m (friction-free). Find its speed at impact.

GIVEN	
EQUATION	
WORK	
ANSWER	

	ME conservation only works when ALL forces are conservative. If friction or air drag acts, you must include their work — which is NEGATIVE.
---	---

Fill in the frame:

The ball's KE at the bottom equals its

_____ at the top, because total
_____ is conserved when only
gravity acts.

EXIT TICKET — DAY 10
A 0.2-kg ball is dropped from 1.8 m. Speed at impact?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can state $KE + PE = \text{const}$ and its condition.	☐	☐	☐	☐
I can apply it to a simple drop and solve for v .	☐	☐	☐	☐
I can identify when ME is NOT conserved (friction).	☐	☐	☐	☐

Day 11 — Conservation — Rolling, Sliding, Swinging

Today's Target

I can apply $KE + PE = \text{const}$ to pendulums, slides, roller coasters, and drops — and recognize that the technique is the same regardless of path shape.

ASTEROID THREAD

What we know about 2026-XJ today: Path SHAPE doesn't change the energy answer — only the heights.

How today connects: That's why we don't need to know the asteroid's exact trajectory. Just its starting and ending heights (effectively, its r values).

DOODLE ZONE

For each station: identify the high-PE and low-PE points. Set ME equal. Solve.

Same rule, four scenarios: $KE_{\text{top}} + PE_{\text{top}} = KE_{\text{bot}} + PE_{\text{bot}}$

1 • PENDULUM



$h_{\text{top}} = \underline{\hspace{1cm}}$
 $v_{\text{bot}} = \underline{\hspace{1cm}}$

2 • SLIDE



$h = \underline{\hspace{1cm}}$
 $v_{\text{bot}} = \underline{\hspace{1cm}}$

3 • ROLLER COASTER 4 • FREE FALL



h_1 vs h_2
 v at top 2 = $\underline{\hspace{1cm}}$



$h = \underline{\hspace{1cm}}$
 $v_{\text{bot}} = \underline{\hspace{1cm}}$

Pattern across all four: v_{bot} only depends on h , not path shape.

Why? $\underline{\hspace{3cm}}$

WORKED EXAMPLE — SLED DOWN A HILL

A 50-kg sled starts at rest at the top of a 12-m hill. Friction-free. Find the speed at the bottom.

GIVEN	$m = 50 \text{ kg} \cdot v_{\text{top}} = 0 \cdot h = 12 \text{ m} \cdot g = 9.81$
EQUATION	$m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2$ (m cancels)
WORK	$9.81 \cdot 12 = \frac{1}{2} \cdot v^2$ $v^2 = 235.4$ $v \approx 15.3 \text{ m/s}$
ANSWER	$\approx 15.3 \text{ m/s}$ — same number for any sled mass.

STATION ROTATION — PENDULUM / COASTER / SLIDE / DROP

At each station, identify the HIGH-PE point and LOW-PE point. Set $ME_{\text{high}} = ME_{\text{low}}$. Solve for the unknown.

STATIONS — FILL IN AS YOU ROTATE	
STATION 1 — PENDULUM: released from $h = 0.6 \text{ m}$ above the bottom. v at the bottom = _____ m/s	
STATION 2 — SLIDE: ramp height 2.0 m , length 5 m , frictionless. v at bottom = _____ m/s	
STATION 3 — ROLLER COASTER: starts from rest at $h_1 = 30 \text{ m}$, then rides to top of next hill at $h_2 = 18 \text{ m}$. v at top of $h_2 =$ _____ m/s	
STATION 4 — FREE FALL: 10-kg rock dropped from 7 m . v at impact = _____ m/s	
★	Across all four: the answer depends ONLY on the HEIGHT difference. Path shape doesn't matter. That's the magic of conservative forces.

Fill in the frame:

Across all four stations, v at the bottom depended only on

_____ — not on the

_____. That's why we say gravity

is _____.

EXIT TICKET — DAY 11

A 0.3-kg pendulum bob is released from 0.6 m above its lowest point (friction-free).
What's its speed at the bottom?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can identify high-PE and low-PE points in any scenario.	☐	☐	☐	☐
I can solve for v using ME conservation.	☐	☐	☐	☐
I can explain why path shape doesn't matter.	☐	☐	☐	☐

Day 12 — Roller Coaster + Pendulum

Today's Target

I can solve a multi-hill roller-coaster problem using ME conservation and find a pendulum's speed at the bottom from its release angle.

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ in an elliptical orbit IS a roller coaster — KE and PE trade off periodically. Faster at perihelion (low U), slower at aphelion (high U).

How today connects: Same equation, different scale. The bookkeeping is identical.

VOCAB · PENDULUM HEIGHT FORMULA

Definition: For a pendulum of length L released from angle θ , the bob's height above its lowest point is $h = L - L \cdot \cos \theta = L(1 - \cos \theta)$.

Example: A 1-m pendulum released at 60° above vertical: $h = 1 \cdot (1 - \cos 60^\circ) = 1 \cdot (1 - 0.5) = 0.5$ m.

Cognates: *Sp. altura de péndulo · Pt. altura do pêndulo*

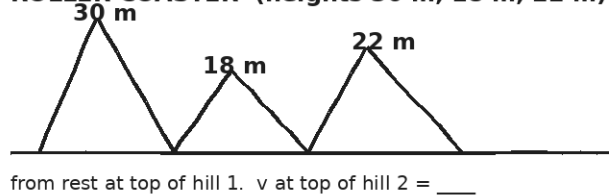
Watch out: Don't use the string length L as the height. Use $L(1 - \cos \theta)$.

DOODLE ZONE

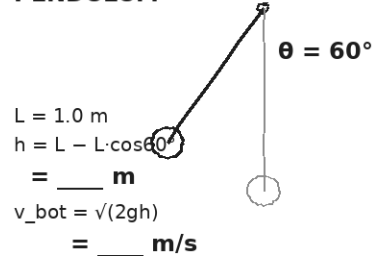
Coaster: find v at top of hill 2 (starts from rest at top of hill 1, $h_1 = 30$, $h_2 = 18$). Pendulum: find h and v_{bottom} for the 60° release.

Two classic scenes — energy bookkeeping

ROLLER COASTER (heights 30 m, 18 m, 22 m)



PENDULUM



Orbital tie-in: perihelion (close to Sun, fast, low PE) $\leftrightarrow$ aphelion (far, slow, high PE)

WORKED EXAMPLE — THREE-HILL ROLLER COASTER

A coaster cart starts from rest at the top of hill 1 ($h_1 = 30 \text{ m}$). Frictionless. Find its speed at the top of hill 2 ($h_2 = 18 \text{ m}$) and hill 3 ($h_3 = 22 \text{ m}$).

GIVEN	$v_1 = 0 \cdot h_1 = 30 \text{ m} \cdot h_2 = 18 \text{ m} \cdot h_3 = 22 \text{ m}$
EQUATION	$ME_1 = ME_2 = ME_3 \rightarrow \frac{1}{2}v^2 + gh = \frac{1}{2}v_1^2 + gh_1 = gh_1$
WORK	At hill 2: $v_2^2 = 2g(h_1 - h_2) = 2 \cdot 9.81 \cdot 12 = 235.4$ $v_2 \approx 15.3 \text{ m/s}$ At hill 3: $v_3^2 = 2g(h_1 - h_3) = 2 \cdot 9.81 \cdot 8 = 157$ $v_3 \approx 12.5 \text{ m/s}$
ANSWER	$v_2 \approx 15.3 \text{ m/s}$, $v_3 \approx 12.5 \text{ m/s}$. Higher hill $\rightarrow$ slower cart.

WORKED EXAMPLE — PENDULUM RELEASED AT 60°

A 1-kg pendulum bob on a 1-m string is released from $\theta = 60^\circ$ above vertical. Find the bob's speed at the bottom of its swing.

GIVEN	$m = 1 \text{ kg} \cdot L = 1 \text{ m} \cdot \theta = 60^\circ \cdot \text{friction-free}$
EQUATION	$h = L(1 - \cos\theta) \quad \text{AND} \quad mgh = \frac{1}{2}mv^2$
WORK	$h = 1 \cdot (1 - \cos 60^\circ) = 1 \cdot (1 - 0.5) = 0.5 \text{ m}$ $v = \sqrt{2gh} = \sqrt{2 \cdot 9.81 \cdot 0.5} = \sqrt{9.81}$ $v \approx 3.13 \text{ m/s}$
ANSWER	$\approx 3.13 \text{ m/s}$ at the bottom of the swing.

YOUR TURN — COASTER BOTTOM FROM 40 M

A coaster cart starts from rest at the top of a 40-m hill. Frictionless. What's its speed at the bottom?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	An asteroid in elliptical orbit: KE and PE trade. Faster near the Sun, slower far away. Same physics as a roller coaster — just on cosmic scale.
---	--

Fill in the frame:

A pendulum at angle θ has height $h =$

_____ above its lowest point. The
bottom speed is $v =$ _____.

EXIT TICKET — DAY 12

A coaster cart starts from rest at the top of a 40-m hill. What's its speed at the bottom?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can solve a multi-hill coaster using ME conservation.	☐	☐	☐	☐
I can find pendulum speed at the bottom from release angle.	☐	☐	☐	☐
I can connect orbital motion (perihelion/aphelion) to coaster physics.	☐	☐	☐	☐

Day 13 — Vernier Ramp Lab — PE → KE · Investigation 4.2

Today's Target

I can release a cart from a measured height, predict v at the bottom from $mgh = \frac{1}{2}mv^2$, measure v with a photogate, and confirm ME conservation across 3 trials.

ASTEROID THREAD

What we know about 2026-XJ today: ME conservation is the rule we'll use Day 19 to convert asteroid PE to impact KE.

How today connects: Today we confirm the rule. If our cart's measured v matches the predicted v , the Day 19 yield calculation rests on solid ground.

LAB SETUP — INVESTIGATION 4.2

Driving question: Does v at the bottom of an incline (predicted from $mgh = \frac{1}{2}mv^2$) match the photogate's measured v , across 3 release heights? Can we recover g from a plot of v^2 vs. h ?

Equipment:

- Vernier dynamics track + cart + tilt apparatus
- Photogate + LabQuest + Logger Pro
- Meter stick + protractor (or pre-measured incline angle)
- Balance for cart mass

What you do:

1. Set incline at $\sim 15^\circ$. Measure incline angle and length.
2. Release cart from $h = 10$ cm (above photogate). Record measured v .
3. Compute predicted $v = \sqrt{2gh}$. Compare to measured. Compute % difference.
4. Repeat at $h = 20$ cm. Then $h = 30$ cm. Three trials total.
5. Plot v_{measured}^2 vs. h . Slope should be $2g$ — does it match?

Safety: Hands clear of cart at the bottom. Catch the cart. Photogate clamped firmly.

How this lab serves the year's question: If our lab confirms PE → KE at small scale, we trust it at asteroid scale on Day 19.

DOODLE ZONE

Sketch your incline + photogate. Record h_1 , h_2 , h_3 as you run trials.

Vernier ramp lab: release from height h , measure v at the bottom.



Predict (no friction):

$$mgh = \frac{1}{2}mv^2$$

$$v_{\text{pred}} = \sqrt{2gh}$$

Trials: $h = 10, 20, 30$ cm

For each, record v_{meas} and % diff.

Plot: v_{meas}^2 vs. h

Slope should be $2g$.

Did you recover g ?

DATA — THREE RELEASE HEIGHTS

Trial	h (m)	v_{pred} (m/s)	v_{meas} (m/s)	% diff	v_{meas}^2 (m ² /s ²)	Notes
1						
2						
3						

OBSERVATION & ANALYSIS

Did $v_{\text{predicted}}$ match v_{measured} ? Within what %? Was the gap larger or smaller at higher h ?

If you plotted v_{meas}^2 vs. h , what was the slope? Did $2g \approx 19.6$ match?

My trials confirmed $mgh = \frac{1}{2}mv^2$ within _____. The biggest source of error was _____. From my slope, I estimated $g \approx$ _____ m/s^2 .

Asteroid connection: how does this serve the year's question?

Fill in the frame:

If ME is conserved, v^2 should be proportional to h with slope $2g$. My slope was _____, implying $g \approx$ _____. Compared to 9.81, my measurement was _____.

EXIT TICKET — DAY 13

Write a CLAIM about what your lab showed, then EVIDENCE (your slope or one % diff), then REASONING (why the slope is approximately $2g$).

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can set up a ramp lab with photogate.	☐	☐	☐	☐
I can compute predicted v from $mgh = \frac{1}{2}mv^2$.	☐	☐	☐	☐
I can compute % difference and infer g from a v^2 vs. h plot.	☐	☐	☐	☐

Day 14 — Hooke's Law: $F_{\text{spring}} = -k \cdot x$

Today's Target

I can write $F_{\text{spring}} = -kx$, compute the spring constant k from a measured stretch + load, and predict the force at any other displacement.

ASTEROID THREAD

What we know about 2026-XJ today: Springs are the textbook template for ELASTIC restoring forces.

How today connects: Spring physics shows up in crumple zones (Unit 3 callback), in atomic bonds, and in any system that wants to return to equilibrium.

VOCAB · HOOKE'S LAW

Definition: The force a spring exerts is proportional to its displacement from equilibrium, in the OPPOSITE direction. $F_{\text{spring}} = -k \cdot x$.

Example: A spring with $k = 200 \text{ N/m}$, compressed 0.08 m : $F = -200 \cdot 0.08 = -16 \text{ N}$ (force pushes BACK toward equilibrium).

Cognates: *Sp. ley de Hooke · Pt. lei de Hooke · HC lwa Hooke*

Watch out: The negative sign is a DIRECTION marker — the force always opposes the displacement. Magnitude is $k \cdot |x|$.

VOCAB · SPRING CONSTANT (K)

Definition: A property of the spring measuring its stiffness. Units: N/m . Bigger k = stiffer spring.

Example: A car suspension spring might have $k \approx 30,000 \text{ N/m}$. A toy slinky might have $k \approx 1 \text{ N/m}$.

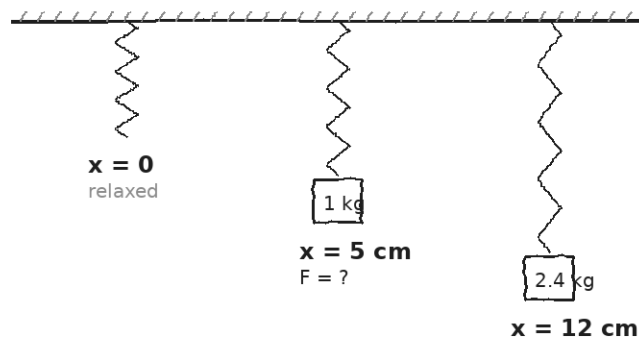
Cognates: *Sp. constante del resorte · Pt. constante da mola*

Watch out: k is a property of the spring, not of the load. Same spring, same k , regardless of how much you stretch it.

DOODLE ZONE

Compute k from case B (1 kg, $x = 5$ cm). Then PREDICT the force in case C (12 cm). Compare to $2.4 \cdot g$.

Hooke's law: $F_{\text{spring}} = -k \cdot x$



Spring constant

from case B:

$$1 \text{ kg} \rightarrow F = m \cdot g = 9.81 \text{ N}$$

$$k = F/x = 9.81/0.05$$

$$k = \underline{\hspace{1cm}} \text{ N/m}$$

Predict case C:

$$F = k \cdot x = (\underline{\hspace{1cm}})(0.12)$$

$$F = \underline{\hspace{1cm}} \text{ N}$$

Compare to $2.4 \cdot g$.

Same? Why?

WORKED EXAMPLE — FINDING K

A spring stretches 5 cm (0.05 m) when a 1-kg mass hangs from it. Find the spring constant. Then: what force does the same spring exert if stretched 12 cm?

GIVEN	$m = 1 \text{ kg} \cdot x_1 = 0.05 \text{ m} \cdot g = 9.81$ Find k . Then F at $x_2 = 0.12 \text{ m}$.
EQUATION	In equilibrium: $F_{\text{spring}} = m \cdot g \rightarrow k \cdot x = m \cdot g \rightarrow k = m \cdot g / x$
WORK	$k = (1)(9.81) / 0.05 = 196.2 \text{ N/m}$ $F \text{ at } x_2 = k \cdot x_2 = 196.2 \cdot 0.12 = 23.5 \text{ N}$
ANSWER	$k \approx 196 \text{ N/m}$. At 12 cm stretch, $F \approx 23.5 \text{ N}$.

YOUR TURN — SPRING UNDER COMPRESSION

A spring with $k = 200 \text{ N/m}$ is compressed 0.08 m . What force does it exert? Which direction?

GIVEN	
EQUATION	
WORK	
ANSWER	

	Hooke's law is a LINEAR approximation. Real springs deviate from it once you stretch them too far ("elastic limit"). For our problems we stay linear.
---	---

Fill in the frame:

Hooke's law says the spring force is PROPORTIONAL to

_____ and OPPOSITE to the
_____ direction. The
proportionality constant is

_____.

EXIT TICKET — DAY 14

A spring with $k = 200 \text{ N/m}$ is compressed 0.08 m . What force does it exert?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute k from a measured load and stretch.	☐	☐	☐	☐
I can predict spring force at any displacement.	☐	☐	☐	☐
I can identify the direction of the restoring force.	☐	☐	☐	☐

Day 15 — Spring Potential Energy: $U_{\text{spring}} = \frac{1}{2} \cdot k \cdot x^2$

Today's Target

I can write $U_{\text{spring}} = \frac{1}{2}kx^2$, compute spring PE for a given compression, and apply $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$ (or $= mgh$) to launch problems.

ASTEROID THREAD

What we know about 2026-XJ today: Spring PE and KE have IDENTICAL mathematical structure: both are $\frac{1}{2} \cdot \text{constant} \cdot \text{squared}$.

How today connects: A spring-launched projectile is a clean model for any device that stores energy and releases it as motion.

VOCAB · SPRING PE

Definition: The energy stored in a compressed or stretched spring: $U_{\text{spring}} = \frac{1}{2} \cdot k \cdot x^2$. Always positive.

Example: A spring with $k = 500 \text{ N/m}$ compressed 0.2 m stores $U = \frac{1}{2} \cdot 500 \cdot 0.04 = 10 \text{ J}$.

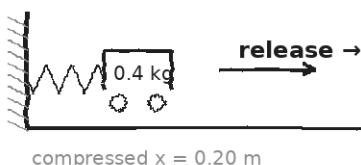
Cognates: *Sp. energía potencial elástica · Pt. energia potencial elástica · HC enèji potansyèl resò*

Watch out: It's $\frac{1}{2}kx^2$, not kx^2 . The $\frac{1}{2}$ comes from averaging force as it grows from 0 to kx during compression.

DOODLE ZONE

Compute U stored in the compressed spring. Set $= \frac{1}{2}mv^2$ and solve for v .

Spring PE $\rightarrow$ KE: $\frac{1}{2} k x^2 = \frac{1}{2} m v^2$



$k = 500$ N/m,
 $x = 0.20$ m,
 $m = 0.4$ kg

PE spring $= \frac{1}{2} k x^2$
 $= \frac{1}{2} \cdot 500 \cdot (0.2)^2$
 $= \underline{\hspace{1cm}} \text{ J}$

$= \frac{1}{2} m v^2 \rightarrow$
 $v = \sqrt{(2 \cdot \text{PE}/m)}$
 $= \underline{\hspace{1cm}} \text{ m/s}$

DERIVATION — WHERE $\frac{1}{2}$ COMES FROM

The spring force grows **LINEARLY** from 0 (at $x = 0$) to $k \cdot x$ (at compression x).

Work = AVERAGE force $\times$ distance. Average force $= \frac{1}{2} \cdot (0 + kx) = \frac{1}{2}kx$.

Therefore: $W = \frac{1}{2}kx \cdot x = \frac{1}{2} \cdot k \cdot x^2$. That stored work IS the spring PE.

WORKED EXAMPLE — SPRING LAUNCHES A CART

A spring ($k = 500$ N/m) is compressed 0.20 m, then released. A 0.4-kg cart sits against the spring, free to slide on a frictionless track. Find the cart's speed when it leaves the spring.

GIVEN	$k = 500$ N/m $\cdot$ $x = 0.20$ m $\cdot$ $m = 0.4$ kg $\cdot$ horizontal, frictionless
EQUATION	$U_{\text{spring}} = \frac{1}{2} \cdot k \cdot x^2 = KE_{\text{cart}} = \frac{1}{2} \cdot m \cdot v^2$
WORK	$U = \frac{1}{2} \cdot 500 \cdot (0.2)^2 = \frac{1}{2} \cdot 500 \cdot 0.04 = 10 \text{ J}$ $\frac{1}{2} \cdot 0.4 \cdot v^2 = 10$ $v^2 = 50$ $v \approx 7.07 \text{ m/s}$
ANSWER	$\approx 7.07 \text{ m/s}$. All 10 J of spring PE became cart KE.

YOUR TURN — VERTICAL LAUNCH

A spring with $k = 800 \text{ N/m}$ is compressed 5 cm (0.05 m). It launches a 0.1-kg ball straight up (friction-free). How high does the ball go?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	$\frac{1}{2}kx^2$ has the SAME structure as $\frac{1}{2}mv^2$. Both are $\frac{1}{2} \cdot (\text{a property}) \cdot (\text{a quantity})^2$. That's not a coincidence — both come from integrating a linearly-growing force.
---	--

Fill in the frame:

A spring launches the cart because the spring's

_____ is fully converted into the
cart's _____. The total energy is
conserved.

EXIT TICKET — DAY 15

A spring ($k = 800 \text{ N/m}$) compressed 5 cm launches a 0.1-kg ball straight up. How high does the ball go (friction-free)?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute $U_{\text{spring}} = \frac{1}{2}kx^2$.	☐	☐	☐	☐
I can set $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$ and solve for v .	☐	☐	☐	☐
I can set $\frac{1}{2}kx^2 = mgh$ and solve for vertical height.	☐	☐	☐	☐

Day 16 — Power: $P = W / t = F \cdot v$

Today's Target

I can compute power as work-per-time, apply $P = F \cdot v$ for instantaneous power, and convert between Watts and horsepower.

ASTEROID THREAD

What we know about 2026-XJ today: Asteroid impact delivers HUGE energy in MICROSECONDS — that's the catastrophic POWER.

How today connects: It's not just the energy. The same energy delivered slowly (a billion years) vs. fast (a millisecond) means lava flow vs. extinction event.

VOCAB · POWER (P)

Definition: The rate of energy transfer. $P = W / t$. Units: Watts ($1 \text{ W} = 1 \text{ J/s}$).

Example: A 700-N person climbs a 4-m staircase in 5 s. $P = (700 \cdot 4) / 5 = 560 \text{ W}$.

Cognates: *Sp. potencia · Pt. potência · HC pisans*

Watch out: Power is NOT energy. Power is the RATE of energy transfer. Big power = fast transfer.

VOCAB · $P = F \cdot V$

Definition: An instantaneous form of power: force times the speed of the point of application. Derived from $P = W/t = F \cdot d/t = F \cdot v$.

Example: A car engine at 75 kW propels the car at 27 m/s. Driving force $F = P/v = 75,000 / 27 \approx 2,780 \text{ N}$.

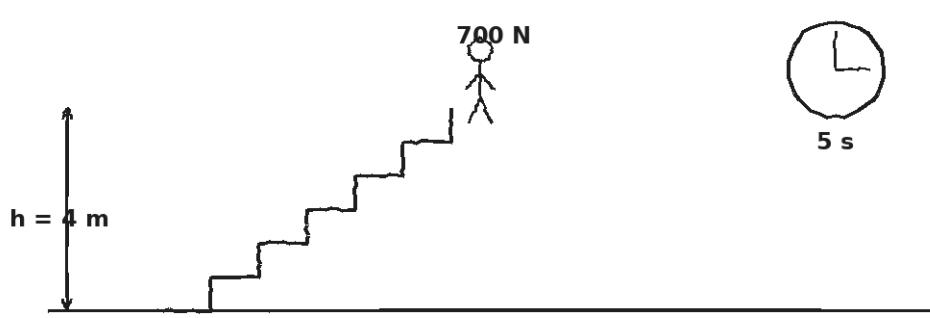
Cognates: *Sp. potencia instantánea · Pt. potência instantânea*

Watch out: Use $P = F \cdot v$ when you know force and speed simultaneously. Use $P = W/t$ when you know work over time.

DOODLE ZONE

Compute W done by the climber. Then divide by 5 s to get power. Compare to a 100-W bulb.

Power = work / time. Watts = J/s.



$W = F \cdot d$
 $= 700 \cdot 4 = \underline{\hspace{1cm}} \text{ J}$

$P = W / t$
 $= \underline{\hspace{1cm}} / 5$
 $= \underline{\hspace{1cm}} \text{ W}$

Compare to a
100 W lightbulb.

Also: $P = F \cdot v$

WORKED EXAMPLE — CLIMBING STAIRS

A 700-N student climbs a 4-m staircase in 5 s. What's their power output?

GIVEN	$F = 700 \text{ N} \cdot d = 4 \text{ m} \cdot t = 5 \text{ s}$
EQUATION	$P = W / t = (F \cdot d) / t$
WORK	$W = 700 \cdot 4 = 2800 \text{ J}$ $P = 2800 / 5 = 560 \text{ W}$
ANSWER	560 W — about 5.6 light-bulbs of power, briefly.

WORKED EXAMPLE — CAR ENGINE POWER

A car engine outputs 100 hp (about 75,000 W) while driving at 27 m/s (≈ 60 mph). What's the driving force?

GIVEN	$P = 75,000 \text{ W} \cdot v = 27 \text{ m/s}$
EQUATION	$P = F \cdot v \rightarrow F = P / v$
WORK	$F = 75,000 / 27 \approx 2,780 \text{ N}$
ANSWER	$\approx 2,780 \text{ N}$ of driving force — balanced by drag + rolling resistance at steady speed.

YOUR TURN — LIFT POWER

A 2-kg ball is lifted 1.5 m in 0.3 s. What power does the lifter generate?

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Asteroid impact: $\sim 10^{17} \text{ J}$ in $\sim 0.1 \text{ s}$. Power $\approx 10^{18} \text{ W}$. The Sun outputs $\sim 4 \times 10^{26} \text{ W}$. For a tenth of a second, an asteroid impact is a millionth-of-the-Sun bright at the impact site.
---	--

Fill in the frame:

Power is the rate of _____

transfer. Two devices can do the same work, but the one with more power does it _____.

EXIT TICKET — DAY 16

A 2-kg ball is lifted 1.5 m in 0.3 s. What power did the lifter generate?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute power as $P = W/t$.	☐	☐	☐	☐
I can apply $P = F \cdot v$.	☐	☐	☐	☐
I can compare powers across human, machine, and natural scales.	☐	☐	☐	☐

Day 17 — KE in Collisions — Elastic vs. Inelastic

Today's Target

I can compute KE_{before} and KE_{after} for an elastic and a perfectly inelastic collision, and explain WHERE the missing KE went.

ASTEROID THREAD

What we know about 2026-XJ today: An asteroid impact is perfectly inelastic. The KE doesn't disappear — it converts to heat, sound, light, mechanical deformation.

How today connects: That's the missing piece: ALL of the asteroid's KE becomes 'damage' in an inelastic event. Day 19 will measure that KE in TNT-equivalents.

VOCAB · KE LOST

Definition: In a perfectly inelastic collision, $KE_{\text{before}} > KE_{\text{after}}$. The DIFFERENCE = the KE converted to other forms (heat, sound, deformation).

Example: 1-kg ball at 6 m/s hits a 1-kg ball at rest, they stick. $KE_{\text{before}} = 18 \text{ J}$. $KE_{\text{after}} = 9 \text{ J}$. Lost: 9 J → heat/sound.

Cognates: *Sp. energía cinética perdida · Pt. energia cinética perdida*

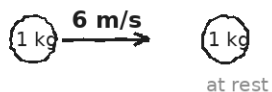
Watch out: "KE was lost" doesn't mean "energy disappeared." Energy is ALWAYS conserved. The KE just transferred to a different form.

DOODLE ZONE

Compute KE_{before} and KE_{after} . Subtract. Where did that energy go? (Squiggles!)

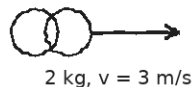
Inelastic collision: KE is NOT conserved. Where does it go?

BEFORE



$$KE_{\text{before}} = \frac{1}{2}(1)(6^2) + 0 = \underline{\hspace{1cm}} \text{ J}$$

AFTER (stuck)



$$KE_{\text{after}} = \frac{1}{2}(2)(3^2) = \underline{\hspace{1cm}} \text{ J}$$

Missing energy = $KE_{\text{before}} - KE_{\text{after}} = \underline{\hspace{1cm}} \text{ J}$
 where did it go?

^^^^

heat / sound / deformation

WORKED EXAMPLE — EQUAL-MASS PERFECTLY INELASTIC

A 1-kg ball at 6 m/s east strikes an identical ball at rest. They stick. Find: (a) v_f from momentum conservation, (b) KE_{before} , (c) KE_{after} , (d) KE lost.

GIVEN	$m_1 = m_2 = 1 \text{ kg} \cdot v_{1i} = 6 \text{ m/s} \cdot v_{2i} = 0$
EQUATION	$m_1 \cdot v_1 = (m_1 + m_2) \cdot v_f$; $KE = \frac{1}{2} \cdot m \cdot v^2$
WORK	$v_f = (1 \cdot 6) / (1 + 1) = 3 \text{ m/s}$ $KE_{\text{before}} = \frac{1}{2} \cdot 1 \cdot 6^2 + 0 = 18 \text{ J}$ $KE_{\text{after}} = \frac{1}{2} \cdot 2 \cdot 3^2 = 9 \text{ J}$ $KE_{\text{lost}} = 18 - 9 = 9 \text{ J}$
ANSWER	Half of the original KE became heat, sound, deformation. The other half stayed as motion.

WORKED EXAMPLE — BULLET INTO BLOCK

A 0.05-kg bullet at 400 m/s embeds in a 5-kg block (ballistic pendulum from Unit 3). Find v_f , KE_{before} , KE_{after} , KE_{lost} .

GIVEN	$m_1 = 0.05 \text{ kg} \cdot v_1 = 400 \text{ m/s} \cdot M = 5 \text{ kg} \cdot V = 0$
EQUATION	$m \cdot v = (m+M) \cdot v_f$; $KE = \frac{1}{2} \cdot m \cdot v^2$
WORK	$v_f = (0.05 \cdot 400) / (5.05) \approx 3.96 \text{ m/s}$ $KE_{\text{before}} = \frac{1}{2} \cdot 0.05 \cdot 400^2 = 4000 \text{ J}$ $KE_{\text{after}} = \frac{1}{2} \cdot 5.05 \cdot 3.96^2 \approx 39.6 \text{ J}$ $KE_{\text{lost}} \approx 3960 \text{ J}$ (over 99% lost!)
ANSWER	Over 99% of the bullet's KE was "lost" — became heat in the wood, deformation, sound.

YOUR TURN — CLAY ONTO STATIONARY BLOCK

A 0.5-kg ball at 10 m/s strikes a 1.5-kg stationary block; they stick. How much KE was lost?

GIVEN	
EQUATION	
WORK	
ANSWER	

	Momentum is conserved in EVERY collision. KE is only conserved in elastic ones. Don't confuse 'energy lost' with 'momentum lost.' Different rules.
---	--

Fill in the frame:

In a perfectly inelastic collision, momentum is

_____ but kinetic energy is
_____. The missing KE becomes
_____.

EXIT TICKET — DAY 17

A 0.5-kg ball at 10 m/s strikes a 1.5-kg stationary block; they stick. How much KE was lost?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute KE_before and KE_after for any collision.	☐	☐	☐	☐
I can quantify the KE lost in a perfectly inelastic collision.	☐	☐	☐	☐
I can name the destinations of "lost" KE.	☐	☐	☐	☐

Day 18 — Heat as the Missing Energy · Bridge to Unit 5

Today's Target

I can state that total energy (KE + PE + thermal + ...) is ALWAYS conserved, and identify thermal energy as the destination of "lost" ME from friction or inelastic collisions.

ASTEROID THREAD

What we know about 2026-XJ today: 2026-XJ's impact KE becomes heat — enough to vaporize rock, raise tsunamis, ignite forests.

How today connects: Unit 5 picks up here. Today's bridge: heat is the form that ME usually becomes when it's "lost."

VOCAB · THERMAL (HEAT) ENERGY

Definition: The energy of random molecular motion. When friction or inelastic collisions "remove" ME, that energy goes into raising the temperature of the objects involved.

Example: Rubbing your hands together: 50 N of friction over 0.5 m → 25 J of thermal energy → hands warm.

Cognates: *Sp. energía térmica · Pt. energia térmica · HC enèji tèmik*

Watch out: Heat is NOT temperature. Heat is the ENERGY that flows; temperature is the AVERAGE motion of molecules.

VOCAB · TOTAL ENERGY CONSERVATION

Definition: The grand principle: SUM of all energy forms (KE + PE + thermal + radiant + sound + chemical + ...) is conserved in any closed system. ALWAYS.

Example: A sled slides down a hill and stops. PE_{top} = thermal generated by friction. (Plus a tiny bit of sound.)

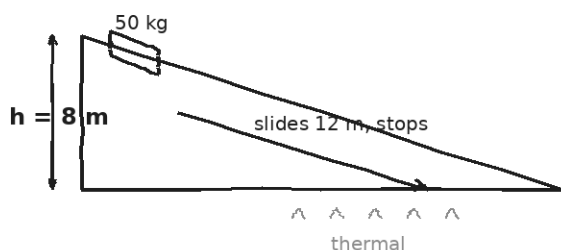
Cognates: Sp. conservación de energía total · Pt. conservação da energia total

Watch out: ME conservation is a SPECIAL CASE — when no friction. Total energy conservation is the broader rule.

DOODLE ZONE

Compute PE at the top. After sled stops at bottom, all of it became thermal. How much?

Friction destroys ME. The lost energy → THERMAL.



$$\begin{aligned} \text{PE}_{\text{top}} &= mgh \\ &= 50 \cdot 9.81 \cdot 8 \\ &= \underline{\hspace{1cm}} \text{ J} \end{aligned}$$

Sled STOPS at bottom:

$$\text{KE}_{\text{end}} = 0, \text{ PE}_{\text{end}} = 0$$

Thermal energy produced = $\underline{\hspace{1cm}}$ J

Energy is conserved
— just in a NEW form.

WORKED EXAMPLE — SLED SLIDES AND STOPS

A 50-kg sled at the top of a hill ($h = 8$ m, friction-free start) slides 12 m down a snowy slope and stops at the bottom (friction at the end zone). How much heat was generated?

GIVEN	$m = 50 \text{ kg} \cdot h = 8 \text{ m} \cdot g = 9.81 \cdot v_{\text{top}} = v_{\text{bot}} = 0$
EQUATION	Total energy: $PE_{\text{top}} + KE_{\text{top}} = KE_{\text{bot}} + PE_{\text{bot}} + Q_{\text{thermal}}$
WORK	$50 \cdot 9.81 \cdot 8 + 0 = 0 + 0 + Q$ $Q = 50 \cdot 9.81 \cdot 8 \approx 3924 \text{ J}$
ANSWER	$\approx 3924 \text{ J}$ of heat generated. All the starting PE became thermal energy.

YOUR TURN — FRICTION ON A HAND-RUB

You rub your hands together. The friction force averages 50 N, you move them 0.5 m of relative distance, back and forth 10 times. How much heat do you generate? (Hint: total distance = $10 \times 0.5 \text{ m} \times 2$ each pass.)

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Unit 5 will tell us how MUCH temperature rise a given heat input produces — and that depends on the material. Steel needs much less heat per °C than water.
---	---

Fill in the frame:

When a sled slides to a stop, the "missing" energy became

_____. Total energy is
 _____ — just in a different
 _____.

EXIT TICKET — DAY 18
A 50-kg sled slides 12 m down a hill (8 m drop) and stops. How much heat was generated by friction?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can state total energy conservation.	☐	☐	☐	☐
I can compute thermal energy generated when an object stops.	☐	☐	☐	☐
I can connect ME loss to thermal gain.	☐	☐	☐	☐

Day 19 — Asteroid Impact Yield — 2026-XJ in TNT Equivalents

Today's Target

I can compute 2026-XJ's KE at impact, convert it to TNT-equivalent tons (1 ton TNT = 4.184×10^9 J), and place the result on the Hiroshima / Tunguska / Chicxulub scale.

ASTEROID THREAD

What we know about 2026-XJ today: Today the entire unit cashes out. We have all the tools: $KE = \frac{1}{2}mv^2$, the asteroid's mass, its impact velocity from Unit 2.

How today connects: Today's number IS the answer to the impact-branch question from UPDATE 4. After today, the question 'how bad is 2026-XJ?' has a real answer in the same units NASA uses for everything else.

VOCAB · TNT YIELD

Definition: The standard unit for comparing energetic events. 1 ton TNT = 4.184×10^9 J. Used for bombs, asteroids, earthquakes.

Example: Hiroshima $\approx$ 15 kilotons = 1.5×10^4 tons. Chicxulub $\approx$ 100 million megatons = 10^{14} tons.

Cognates: *Sp. equivalente en TNT · Pt. equivalente em TNT*

Watch out: "TNT-equivalent" means "this much energy, expressed in the units of how-much-TNT-it-would-take." The actual asteroid is rock, not TNT.

DOODLE ZONE

Compute 2026-XJ's yield in tons. Mark its bar on the log scale. Where does it fall? Closer to Hiroshima? Tunguska? Or much higher?

TNT-equivalent yields — log scale (tons TNT)

1 ton TNT = 4.184×10^9 J

HIROSHIMA

at 1.5×10^4 t

CHELİYABINSK

airburst over Russia 3.0×10^5 t

TUNGUSKA

Siberia airburst, 1908 1.5×10^7 t

2026-XJ (computed)

FILL IN YOUR ANSWER

yield = ____ tons TNT → mark on the axis

CHICXULUB

dinosaur extinction, 66 Ma 1.0×10^{14} t

Where does 2026-XJ fall on this scale? Compute its KE in tons TNT — then draw your own bar

10^4 10^5 10^6 10^7 10^8 10^9 10^{10} 10^{11} 10^{12} 10^{13} 10^{14} 10^{15}

THE BIG COMPUTATION — STEP 1: KE AT IMPACT

2026-XJ: $m = 1 \times 10^9$ kg, impact velocity $v = 3 \times 10^4$ m/s. Compute its kinetic energy at impact.

GIVEN	$m = 1 \times 10^9$ kg · $v = 3 \times 10^4$ m/s
EQUATION	$KE = \frac{1}{2} \cdot m \cdot v^2$
WORK	$KE = \frac{1}{2} \cdot (1 \times 10^9) \cdot (3 \times 10^4)^2$ $KE = \frac{1}{2} \cdot (1 \times 10^9) \cdot (9 \times 10^8)$ $KE = \frac{1}{2} \cdot 9 \times 10^{17}$ $KE = 4.5 \times 10^{17}$ J
ANSWER	$\approx 4.5 \times 10^{17}$ Joules of kinetic energy at impact.

STEP 2: CONVERT TO TNT-EQUIVALENT TONS

Convert your KE to TNT-equivalent tons. 1 ton TNT = 4.184×10^9 J.

GIVEN	KE = 4.5×10^{17} J · 1 ton TNT = 4.184×10^9 J
EQUATION	yield (tons) = KE / (4.184×10^9)
WORK	yield = $(4.5 \times 10^{17}) / (4.184 \times 10^9)$ yield $\approx 1.075 \times 10^8$ tons TNT yield ≈ 108 million tons TNT yield ≈ 108 MEGATONS
ANSWER	$\approx 1.08 \times 10^8$ tons TNT ≈ 108 megatons. (Roughly 7000 Hiroshimas.)

STEP 3: PLACE 2026-XJ ON THE COMPARISON SCALE

WHERE DOES 2026-XJ FALL? (YOUR ANSWER IN CONTEXT)	
• Hiroshima 1.5×10^4 t atomic bomb, 1945 · destroyed a city • Chelyabinsk 5×10^5 t Russia airburst, 2013 · broke windows 100 km out • Tunguska 1.5×10^7 t Siberia airburst, 1908 · flattened 2,000 km² of forest ► 2026-XJ (you) $\approx 10^8$ t your answer · a regional-to-continental event • Chicxulub 1×10^{14} t 66 Ma · killed the dinosaurs	
⚠	The asteroid is fictional. The math is real. NASA's Planetary Defense Coordination Office does exactly this kind of yield calculation for every newly-discovered Near-Earth Object.

DISCUSSION — WHAT DOES $\approx 10^8$ TONS MEAN PHYSICALLY?

DAMAGE SCALE — MATCH TO YOUR COMPUTED YIELD

- 10^4 - 10^5 tons: city-block destruction (Hiroshima-class)
- 10^6 - 10^7 tons: regional flattening (Tunguska-class — 2,000 km²)
- 10^8 - 10^9 tons: multi-continent thermal pulse + months-long climate dimming
- 10^{10} - 10^{12} tons: mass-extinction event begins (smaller K-Pg analog)
- 10^{13} - 10^{14} tons: Chicxulub — 75% of species extinct, dinosaurs gone

Fill in the frame:

2026-XJ would deliver about

_____ tons of TNT-equivalent

energy. That's roughly _____

Hiroshimas, and would correspond to

_____ -scale damage.

EXIT TICKET — DAY 19

Write one sentence stating 2026-XJ's TNT-equivalent yield. Then one sentence comparing it to a historical event. Then one sentence about what that means physically.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can compute the asteroid's KE at impact.	☐	☐	☐	☐
I can convert KE to TNT-equivalent tons.	☐	☐	☐	☐
I can place the asteroid on the historical-event scale.	☐	☐	☐	☐
I can describe in plain language what the yield means physically.	☐	☐	☐	☐

Day 20 — Escape Velocity: $v_{\text{esc}} = \sqrt{(2 \cdot G \cdot M / r)}$

Today's Target

I can derive v_{esc} from $KE = |U|$, compute Earth's escape velocity, and compare it to other bodies (Moon, Mars).

ASTEROID THREAD

What we know about 2026-XJ today: If 2026-XJ had been moving FASTER than the local escape velocity, it could have slipped past Earth (or the Sun) and never come back.

How today connects: Same energy framework as the impact yield — but instead of asking 'how much PE did the asteroid gain falling in,' we ask 'how much KE does it need to climb back out.'

VOCAB · ESCAPE VELOCITY

Definition: The minimum speed at distance r from a body of mass M such that the object can reach infinity with $KE \geq 0$. $v_{\text{esc}} = \sqrt{(2GM/r)}$.

Example: Earth's surface escape velocity: ≈ 11.2 km/s ($\approx 25,000$ mph). Moon's: ≈ 2.4 km/s.

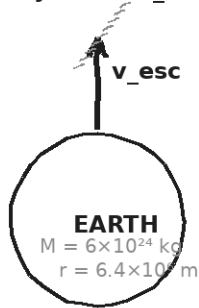
Cognates: *Sp. velocidad de escape · Pt. velocidade de escape · HC vitès chape*

Watch out: Escape velocity doesn't depend on the escaping object's mass. m cancels out.

DOODLE ZONE

Derive v_{esc} by setting $\frac{1}{2}mv^2 = G \cdot M \cdot m/r$ (kinetic energy = magnitude of PE at surface). Solve for v_{esc} .

Escape velocity: $\frac{1}{2} m v_{esc}^2 = G \cdot M \cdot m/r \rightarrow v_{esc} = \sqrt{(2GM/r)}$



$$= 11,200 \text{ m/s}$$
$$= 25,000 \text{ mph}$$

From conservation:

$$KE_{\text{surface}} = |U_{\text{surface}}|$$

$$\frac{1}{2}mv^2 = G \cdot M \cdot m/r$$

$$v_{esc} = \sqrt{(2GM/r)}$$

Plug in:

$$v = \sqrt{(2 \cdot 6.67 \times 10^{-11} \cdot 6 \times 10^{24} / 6.4 \times 10^6)}$$

$$v \approx \text{ ______ } \text{ m/s}$$

$$\approx \text{ ______ } \text{ km/s}$$

What if 2026-XJ had been moving $> v_{esc}$? No impact.

DERIVATION — WHERE v_{esc} COMES FROM

Setup: object at surface (r), mass m , given some launch speed v . At infinity, we want the object to JUST barely have $KE_{\infty} = 0$ (any extra is wasted).

Conservation of total energy: $KE_{\text{surface}} + U_{\text{surface}} = KE_{\infty} + U_{\infty}$

$$\frac{1}{2} \cdot m \cdot v_{esc}^2 + (-G \cdot M \cdot m/r) = 0 + 0$$

$$\frac{1}{2} \cdot v_{esc}^2 = G \cdot M/r \rightarrow v_{esc} = \sqrt{(2 \cdot G \cdot M/r)}$$

Mass m cancels. Escape velocity is a property of the BODY, not the escapee.

WORKED EXAMPLE — EARTH'S ESCAPE VELOCITY

Compute v_{esc} at Earth's surface.

GIVEN	$G = 6.67 \times 10^{-11} \cdot M = 5.97 \times 10^{24} \text{ kg} \cdot r = 6.37 \times 10^6 \text{ m}$
EQUATION	$v_{\text{esc}} = \sqrt{2 \cdot G \cdot M / r}$
WORK	$\begin{aligned} v_{\text{esc}} &= \sqrt{2 \cdot 6.67\text{e-}11 \cdot 5.97\text{e}24 / 6.37\text{e}6} \\ &= \sqrt{2 \cdot 6.67\text{e-}11 \cdot 9.37 \times 10^{17}} \\ &= \sqrt{1.25 \times 10^8} \\ &\approx 1.12 \times 10^4 \text{ m/s} \approx 11.2 \text{ km/s} \end{aligned}$
ANSWER	$\approx 11.2 \text{ km/s} \approx 25,000 \text{ mph}$. Any slower $\rightarrow$ eventually falls back to Earth.

YOUR TURN — MARS ESCAPE VELOCITY

Compute v_{esc} at Mars's surface. $M_{\text{Mars}} = 6.4 \times 10^{23} \text{ kg}$, $r_{\text{Mars}} = 3.39 \times 10^6 \text{ m}$.

GIVEN	
EQUATION	
WORK	
ANSWER	

YOUR TURN — MOON ESCAPE VELOCITY

Compute v_{esc} at the Moon's surface. $M_{\text{Moon}} = 7.35 \times 10^{22} \text{ kg}$, $r_{\text{Moon}} = 1.74 \times 10^6 \text{ m}$.
(This explains why Apollo astronauts could leave the Moon with a small ascent stage.)

GIVEN	
EQUATION	
WORK	
ANSWER	

★	Small bodies (Moon, Pluto, asteroids) have LOW escape velocities. That's why they don't hold atmospheres — gas molecules at typical room temperatures easily exceed v_{esc} and drift away.
---	--

Fill in the frame:

Earth holds an atmosphere because the speed of a typical air molecule at room temperature is _____ Earth's escape velocity. The Moon doesn't because its v_{esc} is _____.

EXIT TICKET — DAY 20

Compute v_{esc} on Mars. $M_{\text{Mars}} = 6.4 \times 10^{23} \text{ kg}$, $r_{\text{Mars}} = 3.39 \times 10^6 \text{ m}$.

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can derive $v_{\text{esc}} = \sqrt{2GM/r}$ from conservation.	☐	☐	☐	☐
I can compute v_{esc} for Earth, Moon, Mars.	☐	☐	☐	☐
I can explain why small bodies don't hold atmospheres.	☐	☐	☐	☐

Day 21 — Three-Branch Energy Budget — Impact / Deflection / Escape

Today's Target

I can run a complete 2026-XJ energy analysis on all three branches (impact yield, deflection feasibility, escape) and articulate which TWO unit ideas were load-bearing.

ASTEROID THREAD

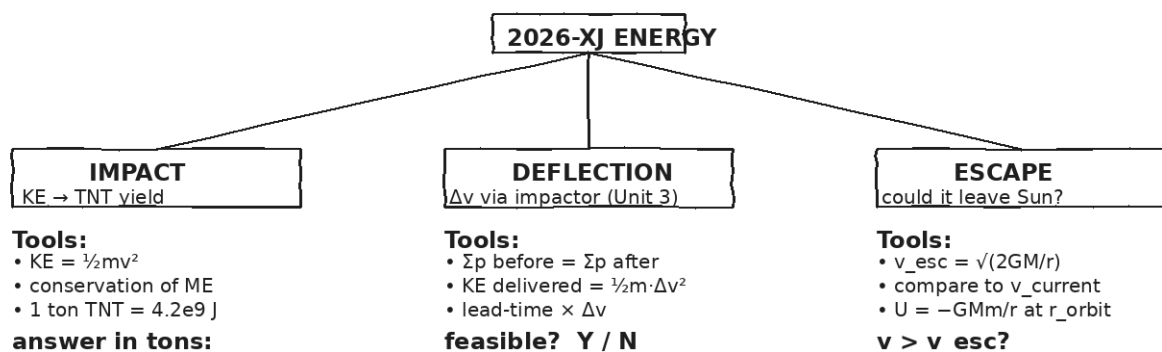
What we know about 2026-XJ today: We have every tool: KE, PE (local + universal), conservation, springs, power, TNT yield, escape velocity.

How today connects: Today is the rehearsal for Day 22's transfer task. Done in groups, compiled as a class, written down in your packet.

DOODLE ZONE

Fill in each branch's **TOOLS** and **ANSWER** box. Then circle **ONE** tool that shows up in **TWO** branches.

2026-XJ — three energy branches



STATION 1 — IMPACT YIELD (REVISIT DAY 19)

Given: 2026-XJ has $m = 1 \times 10^9$ kg, impact velocity $v = 3 \times 10^4$ m/s. (a) Compute KE at impact. (b) Convert to TNT-equivalent tons. (c) Identify it on the Hiroshima/Tunguska/Chicxulub scale.

GIVEN	
EQUATION	
WORK	
ANSWER	

STATION 2 — DEFLECTION ENERGY (REVISIT UNIT 3 DART)

Given: Unit 3 said a single DART-class impactor (600 kg at 6 km/s) delivers about $\Delta v \approx 7 \times 10^{-4}$ m/s to a 5×10^9 kg asteroid. (a) Compute the KE of a single DART spacecraft at launch. (b) That energy translated into Δv on the asteroid — what KE does the asteroid GAIN in its new (slightly faster) state? (Hint: $\Delta KE_{\text{asteroid}} \approx M \cdot v \cdot \Delta v$, where v is asteroid's prior speed.) (c) Comment on whether this is energy-feasible.

GIVEN	
EQUATION	
WORK	
ANSWER	

STATION 3 — ESCAPE FROM THE SUN

Given: 2026-XJ currently orbits the Sun at $v \approx 3 \times 10^4$ m/s (comparable to Earth). Escape velocity from the Sun at Earth's orbital distance ($r = 1.5 \times 10^{11}$ m) is $v_{\text{esc}} \approx \sqrt{2 \cdot G \cdot M_{\text{Sun}} / r}$, with $M_{\text{Sun}} = 2 \times 10^{30}$ kg. (a) Compute v_{esc} . (b) Compare to 2026-XJ's current speed. (c) Could 2026-XJ escape the solar system at its current speed?

GIVEN	
EQUATION	
WORK	
ANSWER	

CLASS CONSENSUS — WHAT'S OUR ENERGY REPORT?

Compile across groups. Write two sentences: one about the impact yield, one about whether deflection AND escape are feasible.

OUR CLASS CONSENSUS — 2026-XJ ENERGY BUDGET	
★	Real planetary defense reports include all three branches. Your numbers are within an order of magnitude of NASA's actual analyses.

Fill in the frame:

Of all the ideas from Unit 4, the TWO most load-bearing for today's analysis were

_____ (used in impact) and

_____ (used in

deflection/escape). Both rest on conservation of

_____.

EXIT TICKET — DAY 21

Of all the things you learned in Unit 4, which TWO ideas were MOST load-bearing for the three-branch analysis? Why?

TODAY'S TARGETS — RATE YOURSELF

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
I can run the impact-yield branch start to finish.	☐	☐	☐	☐
I can run the deflection-energy branch start to finish.	☐	☐	☐	☐
I can run the escape-velocity branch start to finish.	☐	☐	☐	☐
I can articulate which two ideas tied the unit together.	☐	☐	☐	☐

Day 22 — Unit 4 Transfer Task — Assessment Day

ASTEROID THREAD

What we know about 2026-XJ today: Every tool from Unit 4 is in your hands: $W = Fd$ (and $W = Fd \cos\theta$), KE, U (mgh and $-GMm/r$), conservation, springs, power, TNT yield, escape velocity.

How today connects: One of the 5 transfer problems is the asteroid yield problem. Another is escape velocity from a body you've never seen. The rest test technique.

TRANSFER TASK — UNIT 4 · DAY 22

Today you receive the Unit 4 Transfer Task on a separate sheet. It is 5 multi-part problems. You will work independently for ~40 minutes.

- Problem 1 — Work + work-energy theorem. A braking car: given m , v_i , braking force, find the stopping distance using $W_{\text{net}} = \Delta KE$.
- Problem 2 — Conservation of ME. A roller coaster: given heights and a starting speed, find the speed at multiple points. Account for friction in one part.
- Problem 3 — Spring-launched projectile. Given spring constant + compression + launch angle, use $\frac{1}{2}kx^2 \rightarrow KE + (mgh \text{ as needed})$ to find launch speed and max height.
- Problem 4 — Power. An elevator lifting a known mass at a given rate. Compute power required.
- Problem 5 — Asteroid impact yield IN TONS TNT + escape velocity from an unfamiliar body. Asteroid mass + impact speed given; convert KE to TNT tons and place on the scale. Then: compute v_{esc} for a body with given M and r .

HOW THIS WILL BE SCORED — 4 DIMENSIONS, EACH 0-4

Dimension	What a strong (3-4) response shows
Science	Correct physics: vectors, kinematics, units.
Reasoning	Sound method; uncertainty explained, not just stated.
Communication	Work shown, units labeled, organized, readable.
Transfer-to-Asteroid	Honest public-facing connection to 2026-XJ.
MY WORK — DAY 22	

AFTER THE TASK — FINAL UNIT MARZANO (RATE EACH UNIT 4 TARGET)**TODAY'S TARGETS — RATE YOURSELF**

1 = Not Yet (*I can't do this without help*) **2 = Almost** (*I can do this with some help or some errors*) **3 = Got It** (*I can do this independently and explain it*)

Help? Check the box if you want to talk with me about this target — even if you marked yourself a 3.

Target	1	2	3	Help?
K1 — Work as energy transfer · $W = F \cdot d$ and $W = F \cdot d \cdot \cos\theta$.	☐	☐	☐	☐
K2 — Kinetic energy $KE = \frac{1}{2}mv^2$.	☐	☐	☐	☐
K3 — Gravitational PE · mgh (local) and $-GMm/r$ (universal).	☐	☐	☐	☐
K4 — Work-energy theorem $W_{\text{net}} = \Delta KE$.	☐	☐	☐	☐
K5 — Conservation of mechanical energy.	☐	☐	☐	☐
K6 — Power $P = W/t = F \cdot v$.	☐	☐	☐	☐
K7 — Hooke's law and spring PE $\frac{1}{2}kx^2$.	☐	☐	☐	☐
K8 — Thermal energy as the destination of "lost" ME.	☐	☐	☐	☐
S1 — Compute work in 1D and at angles.	☐	☐	☐	☐
S2 — Compute KE for any mass and speed.	☐	☐	☐	☐
S3 — Compute PE near surface and universal.	☐	☐	☐	☐
S4 — Apply $W_{\text{net}} = \Delta KE$.	☐	☐	☐	☐
S5 — Apply ME conservation across scenarios.	☐	☐	☐	☐
S6 — Compute power and energy from rates.	☐	☐	☐	☐

S7 — Apply Hooke's law and spring PE.	☐	☐	☐	☐
S8 — Compute escape velocity.	☐	☐	☐	☐
S9 — Vernier energy measurement (KE lab + ramp lab).	☐	☐	☐	☐
R1 — Reason about sign of work.	☐	☐	☐	☐
R2 — Track energy transformations through collisions.	☐	☐	☐	☐
R4 — Apply energy framework to asteroid impact (TNT yield).	☐	☐	☐	☐
R5 — Reason about escape velocity using energy conservation.	☐	☐	☐	☐

Equation Reference — Unit 4

Keep this page open while you work. Every equation below is on the MCAS Equation Sheet — you do NOT need to memorize them. You DO need to recognize when each one applies.

Concept	Equation	When to use it
Work (force along motion)	$W = F \cdot d$	Units: Joules (N·m). For force directly along the direction of motion. Carrying a book level = zero work (force perpendicular to motion).
Work at an angle	$W = F \cdot d \cdot \cos \theta$	θ = angle between F and d. $\theta = 0 \rightarrow$ full work. $\theta = 90^\circ \rightarrow$ zero. $\theta > 90^\circ \rightarrow$ negative work (friction).
Kinetic energy	$KE = \frac{1}{2} \cdot m \cdot v^2$	Units: Joules. The squared v is what makes KE non-linear in speed. Double v $\rightarrow 4 \times$ KE. The asteroid yield calculation depends on this.
Work-energy theorem	$W_{\text{net}} = \Delta KE$	Net work done on an object equals change in its KE. Skip kinematics — go straight from forces to final speed.
Gravitational PE (near surface)	$U = m \cdot g \cdot h$	Near Earth's surface ($g \approx 9.81$). Reference height matters — only DIFFERENCES in PE are physical.
Gravitational PE (universal)	$U = -G \cdot M \cdot m / r$	For any altitude. PE = 0 at infinity, negative everywhere closer. Use when h is comparable to Earth's radius (asteroid distances).
Conservation of ME	$KE + PE = \text{constant}$	Holds when only conservative forces act (gravity, springs). Friction breaks this — that energy becomes thermal.
Hooke's law	$F_{\text{spring}} = -k \cdot x$	Spring force, where k = spring constant (N/m), x = displacement from equilibrium. Negative because force restores toward equilibrium.
Spring PE	$U_{\text{spring}} = \frac{1}{2} \cdot k \cdot x^2$	Energy stored in a compressed/stretched spring. Area under F-vs-x curve (triangle, $\frac{1}{2} \cdot \text{base} \cdot \text{height}$).
Power	$P = W / t = F \cdot v$	Rate of energy transfer. Units: Watts (J/s). 1 hp ≈ 746 W. Instantaneous form: $P = F \cdot v$.

Escape velocity	$v_{\text{esc}} = \sqrt{2 \cdot G \cdot M / r}$	Speed needed at distance r from a body of mass M to escape its gravity well (reach infinity with $v \geq 0$). Earth: ≈ 11.2 km/s.
TNT-equivalent conversion	1 ton TNT = 4.184×10^9 J	Divide an event's KE in Joules by this to get its TNT yield in tons. The unit Hiroshima, Tunguska, Chicxulub are reported in.

Constants and reference quantities:

$G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$ · $M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$ · $R_{\text{Earth}} = 6.37 \times 10^6 \text{ m}$ (6,371 km) ·

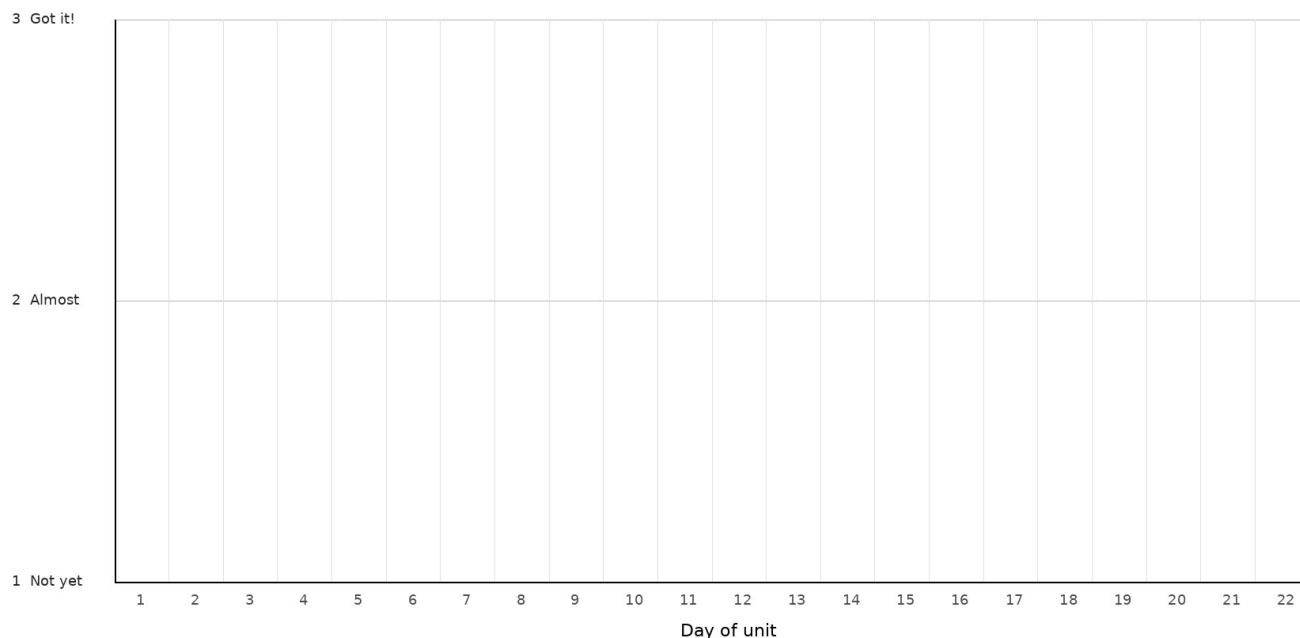
$g_{\text{surface}} \approx 9.81 \text{ m/s}^2$ · 1 ton TNT = $4.184 \times 10^9 \text{ J}$ · 1 hp $\approx 746 \text{ W}$ · 2026-XJ: $m \approx 10^9 \text{ kg}$, $v_{\text{impact}} \approx 3 \times 10^4 \text{ m/s}$

TNT-EQUIVALENT YIELDS — FOR COMPARISON ON DAY 19

Event	TNT-equivalent yield	Context
Hiroshima	1.5×10^4 tons (15 kilotons)	Atomic bomb, August 1945. Destroyed a city.
Chelyabinsk	5×10^5 tons (0.5 megaton)	Russia, 2013. Airburst. Shattered windows for 100+ km.
Tunguska	1.5×10^7 tons (15 megatons)	Siberia, 1908. Flattened 2,000 km ² of forest.
2026-XJ (compute)	??? tons (your Day 19 answer)	Your job: place this on the scale.
Chicxulub	1×10^{14} tons (100 million Mt)	66 million years ago. Killed the dinosaurs.

Mastery Growth Chart — Unit 4

Each day, transfer your Marzano 1-2-3 rating onto the chart. Use a different mark or color for each target you track. Unit 4 has the most targets (15+) of any unit so far — track at least your top three.



LEGEND — WHICH MARK / COLOR = WHICH TARGET

Mark: _____ Target: _____

Mark: _____ Target: _____

Mark: _____ Target: _____

Mark: _____ Target: _____

NOTES — END-OF-UNIT REFLECTION